Gendered Misallocation of Agrarian Labor*

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October 5, 2026

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Abstract

We test a novel explanation for disparities in agricultural output between men and women in Malawi: child-bearing as a unique shock to women's labor supply that inhibits efficient land utilization. We implement a randomized controlled trial with female farmers who are pregnant or have a child under the age of one years old. We test the impact of providing five paid days of skilled agricultural labor during the farming season. For women who are enrolled in the trial during the first two trimesters of pregnancy, the program does not substitute for household labor, and, if anything, crowds *in* hired labor. These treated women experience a 0.26 standard deviation increase in harvest yields and a 0.39 standard deviation increase in maternal food security. For women enrolled in the trial during the third trimester of pregnancy, the treatment leads farmers to substitute away from hired labor and engage more in non-agricultural income-generating activities, with no effects on yields or food security. The program has no impact among women who are non-pregnant. We interpret these results through a model where the physiological labor supply shock of pregnancy interacts non-linearly with market failures and intra-household norms to create gendered shocks to farm productivity and consumption.

*Thank you to One Acre Fund Malawi and Invest in Knowledge Initiative, without whom this project would not be possible. Special thanks to Yash Gharat, Laetitia Pham, Nandi Bwanali, Emmanuel Malata, Ojas O, Fanny Kulisewa, Flora Magaleta, and James Mkandawire, who were instrumental in the intervention design, implementation, and data collection. We are thankful for useful comments and suggestions from Sarah Baum, Emily Breza, Margaret McConnell, Jessica Cohen, and numerous conference and seminar participants. We gratefully acknowledge financial support from the Weiss Fund at the University of Chicago and the John C. Hansen & Katherine Vogelheim Research and Travel Fund for Africa. This study received IRB approval from Harvard University Area, the Harvard Longwood Campus, and the Malawi National Committee on Research in Social Sciences and Humanities. The experiment was pre-registered under the AEA RCT Registry (AEARCTR-0014981) and ClinicalTrials.gov (NCT06631768).

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1 Introduction

Female-managed farming plots in low- and middle-income countries (LMICs) systemically produce less than their male counterparts (Bambio, 2025; Oseni et al., 2014; Pierotti et al., 2022; Udry, 1996). Existing evidence finds that the disparity in output is driven by disproportionate allocation of resources: women receive less or worse seeds, land, fertilizer, and labor (Akram-Lodhi, 2018; Burke and Jayne, 2021; UN Women, 2015). The primary source of labor for smallholder farmers is the farm owner herself. However, female farmers experience a double burden of labor in which they are expected to split their time between domestic responsibilities and income-generating activities. While substituting for female farmer's domestic labor with paid childcare can improve farm productivity (Donald, 2024), it is impossible to substitute for women's time and energy in some elements of domestic work, including pregnancy and postpartum recovery. Moreover, the reduction in labor time and intensity caused by pregnancy is not adequately replaced by the labor of male household members (Bryson et al., 2021). Does the labor productivity shock represented by pregnancy in physically intensive work contribute to low productivity of female-managed plots? Do households reallocate land and labor inputs effectively to manage this labor productivity shock?

First, we quantify the bidirectional relationship between fertility behavior and agriculture. Using Malawi Demographic and Health Survey data, we find that births are most common in the height of the agricultural season. This implies that women experience pregnancy, a physiological phenomenon that increases nutritional requirements and makes physical activity more challenging, precisely at the same time that labor demand is highest and food prices are highest. Using data from One Acre Fund Malawi (OAF), a non-profit agricultural organization, we find that women who are pregnant for most of the farming season report working 15% fewer hours per week than non-pregnant women. While prior studies in sub-Saharan Africa have explored interventions to alleviate the constraints women face in allocating time between domestic responsibilities and income-generating activities, these interventions often focus on reducing the burden of childcare and tend to target those women whose youngest child is aged 1 year or older.[1] For women who are pregnant or postpartum, it may not be optimal or possible to reduce physiological and childcare barriers to engaging in agricultural employment without causing harm to maternal and child health.

Next, we conduct a cluster-randomized controlled trial in Malawi to test if labor short-

[1]There is evidence from across national contexts in sub-Saharan Africa that suggests that improving access to childcare resources increases maternal employment and income (Ajayi et al., 2022; Bjorvatn et al., 2025; Clark et al., 2019; Donald, 2024). In particular, Donald (2024) find that establishing childcare centers in rural communities in the Democratic Republic of the Congo increases the probability of female participation in farming crops for sale by 11 percentage points and increases the overall household production of produce via improving the productivity of female-managed plots.

ages during pregnancy contribute to low farm productivity. We partner with OAF to provide subsidized casual farm labor, sourced and trained by OAF, to pregnant women and women caring for infants (the Planting Partners program). OAF is a non-profit organization that provides training and agricultural inputs to farmers on credit, such as seeds and fertilizers. Consequently, this sample is land-owning farmers with access to complementary capital inputs, but who typically cultivate their land with their own labor supply. The intervention effectively resolves myriad frictions to hiring farm labor (Breza and Kaur, 2025), including credit market frictions (Feder, 1985; Eswaran and Kotwal, 1986; Karlan et al., 2014), search frictions (Foster and Rosenzweig, 2022), labor supply rationing (Breza et al., 2021; Dillon et al., 2019; Kaur, 2019), and quality control (Frisvold, 1994; Foster and Rosenzweig, 1994).

The treatment leads to substantial effects on farm yields (0.26 SD, $p < 0.10$) and maternal food security (0.39 SD, $p < 0.05$) among women who enter the trial early in their pregnancy. Solid food consumption for the youngest child decreases by 0.24 SD ($p < 0.05$), without any increases in the frequency of breastfeeding. However, this is likely driven by the age of the youngest child in this group and feeding recommendations.[2] While these women's pregnancies overlap with most of the farming season, they have the ability to provide some on-farm labor for much of the season and many do not give birth until after the harvest. Women who are in their third trimester at the start of the farming season do not experience any treatment effects on production or consumption. These women all give birth in the middle of the agricultural season, and thus are severely constrained in their own ability to supply labor.[3] There are no treatment effects among women who are not pregnant but caring for infants, implying that the binding labor supply constraint is likely not driven by time dedicated to childcare. Jointly, these results suggest that the physiological constraint of pregnancy limits women's ability to efficiently supply labor to their farm. The Planting Partners program was sufficient to overcome the labor supply shock of early- and mid-term pregnancy, but not enough to overcome the labor supply shock of spending most of the agricultural season in late-term pregnancy and postpartum recovery.

We propose a conceptual framework that extends standard agricultural household models and models of separation failures to explain these results (de Janvry et al., 1991; Singh et al., 1986; Benjamin, 1992). Consistent with prior models and empirical evidence demonstrating complementarities between farm labor supply and other farm inputs (Fink et

[2]For 75% of women in this group, the youngest child is less than 6 months old, the age up until which the World Health Organization recommends exclusive breast feeding (World Health Organization, 2026). All are less than 8 months old.

[3]In addition to experiencing childbirth during the phase of the season where the timing of labor supply has large implications for yields, they spend most of the agricultural season in late-term pregnancy or postpartum recovery.

al., 2020; Feder, 1985; Eswaran and Kotwal, 1986), we allow for a non-convexity in the marginal returns on labor. We assume that households always supply enough farm labor to produce food for subsistence, even when returns are lower than off-farm wages. In a standard model with complete markets, while pregnancy leads to a reduction in total household labor supply available, it should not affect the productivity of *land*. Households can reallocate on-farm and off-farm labor between household members, can hire labor to cultivate the land, or can rent land out. We test if these margins of substitution are adequately available in the face of a labor productivity shock by estimating treatment effects on land use and labor allocations.

If pregnant control women are able to effectively utilize land renters and labor substitutes for their own farm labor, the Planting Partners program should crowd out at least one of these margins of substitution. Among women in all stages of pregnancy, there is no reduction in on-farm labor supply from other household members, suggesting there are constraints to intra-household labor substitution.[4] For women who enter the trial early in pregnancy, the treatment crowds *in* hired labor, while for women who enter the trial late in pregnancy, the treatment crowds *out* hired labor. There is no effect among women who are not pregnant. We argue that women who enter the trial late in pregnancy are unable to supply enough non-Planting Partners labor to benefit from labor complementarities, and choose to supply enough labor for subsistence with or without Planting Partners. Consistent with this hypothesis, we observe treated women who enter the trial late in pregnancy are more likely to engage in non-agricultural income-generating activities, suggesting they reinvest savings from hired labor into more profitable non-agricultural activities. Women who enter the trial early in pregnancy are able to supply labor productively enough so that, together with Planting Partners labor inputs, they are able to access complementarities, use land efficiently, and produce greater yields.

Jointly, this evidence demonstrates that pregnancy generates a gender-specific labor productivity shock that households are unable to smooth over, given intra-household norms and market failures. Supplying insufficient labor so as to access labor complementarities further amplifies the economic consequences of this labor supply shock. As a result, women experience significantly greater consequences of the features and market failures that are characteristic of developing agricultural economies.

This paper contributes to the literature on agricultural productivity among smallholder farmers in developing countries. We show that pregnancy, an almost universal yet gendered household labor productivity shock in agricultural societies, exacerbates the implications of multiple market failures. Conceptually, we extend standard models of agricul-

[4]There is likewise no reduction in labor hours supplied by the farmer herself, but this may mask increased time acting in a supervisory role while decreasing physically intensive work that hired labor can do more efficiently than pregnant women.

tural households to consider the implications of separation failures for gendered productivity differences when women systematically face a labor productivity shock (Benjamin, 1992; de Janvry et al., 1991; Dillon et al., 2019; Jacoby, 1993; Jones et al., 2022; LaFave and Thomas, 2016; Singh et al., 1986).[5] Empirically, we demonstrate that constraints on self-sourced labor supply can have significant implications for agricultural farm productivity, but that a simple intervention that makes hired labor accessible can mitigate productivity losses. While there is extensive evidence on the importance of hired farm labor as an available form of employment for agricultural households (Benson and De Weerdt, 2023; De Weerdt et al., 2026; Fink et al., 2020), there is less evidence on demand for casual hired labor, and, to our knowledge, no other paper that directly experiments with providing it.[6]

Second, this paper contributes to the literature on the economic empowerment and health of women in agricultural societies. A large body of work shows that women face systematic constraints in agricultural settings, including limited access to inputs, labor, and decision-making authority (Doss, 2018, 2013; Doss and Morris, 2001; Doss et al., 2015; Udry, 1996). We show that the economic consequences of these limitations on intra-household substitution of agricultural inputs is exacerbated by pregnancy. At the same time, resolving labor market frictions has substantive impacts on women's food security. These findings have direct policy implications for public health, as maternal and child malnutrition continue to be pressing public health issues in many low- and middle-income countries (LMICs), which directly impact people's well-being, survival, and productivity (Almond et al., 2018; Hoddinott et al., 2008; Maccini and Yang, 2009; Victora et al., 2021). Our program delivers large gains in maternal food security cost-effectively, with suggestive evidence of downstream benefits for breastfeeding infants. Comparing our estimates with maternal cash transfers in India (Weaver et al., 2024), our program delivers very similar treatment effects on maternal food security at one-tenth the cost.

Third, the paper contributes to the broader literature on gender, labor supply, and the economic consequences of childbearing. Much of the existing literature, particularly in

[5] Palacios-López and López (2015) develop a conceptual argument with a similar result whereby market failures will impact female-managed farm plots more than male-managed farm plots. Their underlying mechanism differs from ours: they argue that women's off-farm labor is lower-return than men's (for example, because of lower wages or higher commuting costs), leading women to allocate more labor to the farm than men. Off-farm work offers liquidity, so women are more exposed to market failures than men. While we do not test or make any claims about gendered differences in the consequences of market failures outside of when women are pregnant, note that these forces can also operate in our setting; our argument is that, if these differential responses to market failures already exist, pregnancy will deepen them further.

[6] While there is an extensive theoretical literature on the implications of labor market frictions in agricultural economies, much of the causal literature has tested smoothing other market frictions by providing credit, insurance, or capital/new technologies (Fink et al., 2020; Karlan et al., 2014; Emerick et al., 2016; Caunedo and Kala, 2021). Several experiments in urban settings directly alleviate labor market frictions within firms in developing countries (de Mel et al., 2019; Hardy and McCasland, 2023).

high-income contexts, emphasizes the role of labor supply decisions, employer responses, and institutional factors in generating gender gaps following childbirth (Bertrand et al., 2010; Kleven et al., 2019a,b; Angelov et al., 2016). In contrast, we show that in settings where work is physically intensive and markets are incomplete, the biological component of pregnancy itself can generate meaningful productivity losses. These results suggest that the economic impact of childbearing depends not only on labor market institutions and norms, but also on the physicality of work and the extent to which households can access substitute inputs.

In Section 2, we describe the experimental design and data. Next, in Section 3, we describe the context, and present descriptive evidence on the seasonal relationship between fertility, agricultural production, and labor demand. Section 4 presents the treatment effects of the intervention on our main downstream outcomes: farm yields, maternal food security, and child nutrition. Section 5 presents a conceptual framework of how the gendered labor productivity shock of pregnancy and incomplete markets can interact to affect agricultural household decisions; guided by the framework, Section 6 estimates treatment effects on labor and land allocations. Finally, Section 7 concludes.

2 Experimental Design

We partner with One Acre Fund Malawi, a non-profit organization that provides training and agricultural inputs on credit to farmers, to design and implement the Planting Partners program. While OAF selects households who own land and provides them access to agricultural capital inputs, its model relies on farmers being able to supply their own labor to their farms or hire farm labor themselves. While most farmers are likely selected on the ability to supply farm labor, pregnancy represents a common but intense physiological shock to women's labor productivity. Planting Partners was designed to fill this gap and help child-bearing female farmers mitigate adverse consequences of labor supply shocks to their farm's productivity.

Site Selection and Eligibility

We conduct the RCT across four districts in Malawi: Chiradzulu, Dedza, Lilongwe, and Machinga, which span both the Central and Southern regions of Malawi. We randomize the intervention at the "site" level, the geographic unit that OAF uses to administer services such as trainings and farmer monitoring. Each site includes several villages, but is smaller than any governmental administrative units. There are approximately 200 OAF sites across the four study districts. We randomly select 75 sites for participation in the trial, stratified at the district level to provide representativeness across multiple

regions that OAF operates in.[7] From these 75 sites, we randomly select 38 as treatment sites and use the remaining 37 as control sites.

Women were eligible within these sites if they were enrolled OAF farmers, and if they were pregnant or had a child under 1 year old in their household at the time of recruitment (November 2024). Among ten sites that had more than 45 eligible women, 45 eligible women were randomly selected from each site. In the remaining sites, all eligible women were included (on average, each site had 26 eligible women).

Planting Partners Program Details

OAF screened, hired, and trained between two and three casual workers per site, depending on the number of eligible women. Each casual worker was provided with a hoe from OAF. Workers were paid 5,500 Malawi Kwacha (3.14 USD) per day of work for labor and to cover transportation costs to recipient farmers' plots, with an additional 1,000 MWK (0.57 USD) lump sum for the entire program duration for airtime. Selected women in treatment sites were offered a voucher that they could redeem up to five times, with each redemption equating to a day of work supplied by the OAF casual worker hired within that site. Per protocol, workdays were meant to consist of six hours of work and a one-hour break. Farmers were told by OAF that they must also be present at the farm when the casual worker is there to provide oversight.

Labor from casual workers provided by the program could only be used to assist with farming-related activities associated with the OAF-sponsored plot managed by each beneficiary household. An OAF-sponsored plot is one in which the farmer uses OAF inputs, including seeds, fertilizer, and OAF-encouraged farming methods. These OAF-sponsored plots are primarily for the purpose of cultivating maize, but there are instances where households intercrop these plots with soybeans. Initially, Planting Partners was intended to run from the planting period to the end of weeding, just before harvest. However, the program was extended midway through to include harvest and post-harvest land clearing. Farmers in both treated and control sites continued receiving the standard OAF programming.

Randomization was conducted in November 2024, and implementation of the "Planting Partners" program began in December 2024. We conducted a survey to measure impacts of the intervention after the conclusion of the harvest season in August and September of 2025 using a third-party survey firm. Details of this trial and analysis were recorded before program conclusion on ClinicalTrials.gov (NCT06631768) and AEA RCT Registry (AEARCTR-0014981).

[7] The Central and Southern regions in Malawi are distinct culturally and agriculturally.

2.1 Data

OAF conducted a survey of enrolled female farmers to determine eligibility, collecting basic information on pregnancy status, household members and their respective sexes and ages, and self-reported measures of productivity. Participants estimated the daily amount of hours they work and the lines they complete in a day, which serve as measures of both time and efficiency of farm labor. Farmers were unaware of the Planting Partners program at this time, and surveyors were blinded to treatment assignment of sites. During the endline survey, data were collected on household characteristics, household crop production, farm labor quantity and sources, fertility history, maternal and child food consumption, and social networks.

2.1.1 Outcomes

Our primary outcomes of interest for this trial are total harvest yield, measured in US dollars (USD), maternal food security, and child nutrition.[8]

Total Harvest Yields: For each crop that a farmer produced, they report their harvest yields in the unit that they collect or store that crop. We convert their yields into kilograms using OAF conversion charts by crop and unit. We ask each farmer the price per unit that they sold each crop (estimated if they have not yet sold any), and create a district-level price for each crop using the median expected price reported in control sites in each district. The total value of each crop is then winsorized at the 1st and 99th percentile and then summed across every crop.

Maternal Food Security: We construct a standardized measure of food security comprised of farmer's responses to questions regarding how many times they went to bed hungry over the past month (0 = never; 1 = a few times or more); frequency of meals the day before and two days before (0 = one or two meals; 1 = three or more meals); and the numbers of days in the past month they consumed meat, egg, *usipa* (a preserved fish foodstuff), and fish.

Child Nutrition: We similarly construct a standardized measure of solid food consumption comprised of the reported number of days in the past month the child had meat; fruits and vegetables; and carbohydrates such as porridge, bread, rice, pap, or pasta. Because some of the farmers in this study gave birth during the agricultural season and thus their youngest child would not be expected to have transitioned to solid food, we also measure treatment effects on average daily frequency of breastfeeding of the youngest child when recalling over the past week.[9]

[8] Data on crop prices were collected in Malawi Kwacha, and then converted to USD using the official conversion rate from 1 August, 2025 (1 USD = 1,751 MWK).

[9] Comparable measures to our maternal food security and child nutrition outcomes have been employed by prior studies, including Croke et al. (2022); Tesfay (2020); World Food Programme (2025).

2.1.2 Sample Characteristics and Program Implementation

Data Collection Timeline

Using administrative data from 13,585 female farmers across the 75 study sites, we determine that 1,851 farmers are eligible. After eligibility is assessed and farmers are selected for the program, we randomize sites to treatment and control. There was concern that eligibility was not always accurately assessed in the initial survey, so OAF conducted a verification process in treated areas in November 2024, to ensure the program was only administered to eligible farmers. Four sites were dropped from the trial entirely, because there were too few eligible women for One Acre Fund to be able to logistically provide the Planting Partners intervention. Given time constraints and the need to conduct this verification prior to the program roll-out, OAF only conducted the verification process in treatment areas. They conducted a similar verification process in control areas in April 2025. We hired a third-party survey firm, Invest in Knowledge Initiative, to collect endline data in August-September 2025, before the following agricultural season.

Analytical Sample Criteria

The verification process may have created a differential selection process for the endline survey, since verification surveys were conducted after treatment status was assigned, and at different times in treatment and control areas. To mitigate concerns about differential selection, we restrict our analytical sample to farmers whose eligibility is determined using endline data collected by a third-party survey firm. In the endline survey, households report the ages of each household member, the date of the last time they gave birth, and if they have had a miscarriage since November 2024. These variables allow us to impute eligibility (pregnancy or having a household member under 1 year old) in November 2024. This restriction drops 134 households, non-differentially between treatment and control.

We drop another 46 households (16 control and 30 treated) whose name did not match our originally recorded sample from the administrative data we used to determine eligibility. In some cases, these are situations where a treated respondent transferred the Planting Partners vouchers to somebody else. In other cases, in both treatment and control areas, farmers registered with OAF under a family member's name. While the second case does not create issues of differential selection across treated and control areas, we remove all households where the names do not align to be as conservative as possible and eliminate all threats to treatment imbalance. Ultimately, we drop 16% of control respondents and 20% of treated respondents. Our final analytical sample consists of 440 farmers in the treated group and 368 farmers in the control group.

This process allows us to rule out differential inclusion of ineligible women across treatment and control sites. This means that we do not have to worry about favoritism

Table 1: Balance of characteristics between treatment and control groups

Characteristic	Control Farmers (N = 359)	Treated Farmers (N = 433)	p-value	N	Clusters
Proportion reported pregnant at time trial start in OAF data	0.53 (0.03)	0.48 (0.02)	0.319	792	71
Reported household size at trial start	4.53 (0.09)	4.96 (0.08)	0.013	792	71
Reported lines worked per day	22.36 (0.79)	25.86 (0.69)	0.445	792	71
Reported hours worked per day	4.22 (0.10)	4.10 (0.06)	0.559	792	71
Married and living together	0.82 (0.02)	0.83 (0.02)	0.837	792	71
Number of prior births	3.38 (0.09)	3.27 (0.09)	0.516	792	71
Age	32.63 (0.34)	32.83 (0.37)	0.376	792	71
Proportion reported pregnant during trial in research data	0.47 (0.03)	0.41 (0.02)	0.104	792	71
Total acres owned	1.65 (0.06)	1.83 (0.06)	0.214	791	71
F-test of joint significance (F-stat)		1.47	0.176	791	71

Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

towards ineligible women in treatment sites, which is our greatest concern with the post-treatment-assignment verification process. While we cannot decisively rule out differential exclusion of different types of people in treatment and control sites, we deploy extensive balance checks within included and excluded groups. Furthermore, it is difficult to rationalize why the post-treatment verification process would lead to differential exclusion, since participants want to be included in a program that provides access to a free service.

Treatment Balance

The final sample is balanced overall between treated and control groups ($p = 0.176$ for joint test, Table 1).[10] There is a statistically significant but quantitatively small imbalance between the two groups on household size reported in administrative data. Treated households are slightly larger, with 4.96 versus 4.54 household members ($p = 0.013$). Reassuringly, household size is uncorrelated with total yields in the control group. Furthermore, when we evaluate the sample of participants who are dropped from the trial (Table A.9), women in treatment and control sites are balanced on all variables except for household size, where treatment women again have larger households (5.06 household members versus 4.5 household members, $p = 0.068$). This suggests that differential

[10]The first four variables of Table 1 are from the administrative data collected prior to treatment assignment. The next five variables are variables that we collect after the intervention with our third party data collection firm. We include these five measures for balance since they either cannot be affected by the treatment (number of births prior to the intervention, age, and pregnancy status at the start of the trial), or are very unlikely to be affected by the intervention (marital status and the total number of acres owned).

household size is due to imbalance in randomization, which we can control for, rather than differential selection into our analytical sample.

Since heterogeneity by pregnancy status is important to our research question and results, we also evaluate treatment balance within sub-groups: women who were never pregnant during the trial, women who entered the trial in the first two trimesters of pregnancy, and women who entered the trial in the third trimester of pregnancy. Unsurprisingly, given smaller sample sizes, there is some imbalance within these sub-groups, and three variables emerge as unbalanced in these sub-group analyses: married and cohabitating with a spouse (negatively associated with treatment status among non-pregnant women, but positively associated with treatment status among pregnant women in any stage of pregnancy); age (negatively associated with treatment status among women who enter the trial early in pregnancy); and the total acres of land owned (positively associated with treatment status among women who enter the trial late in pregnancy). Reassuringly, the directions of these margins of imbalance are unlikely to drive our heterogeneous results. Imbalance on age in particular is a concern, because it is the only variable that is unbalanced only within the early-pregnancy group, the only group where we see treatment effects. However, while the imbalance is statistically significantly, the quantitative difference in age between treatment and control in this group is small (32 years old on average among control, versus 30 years old on average among treatment), and age is unassociated with yields in the control group.[11]

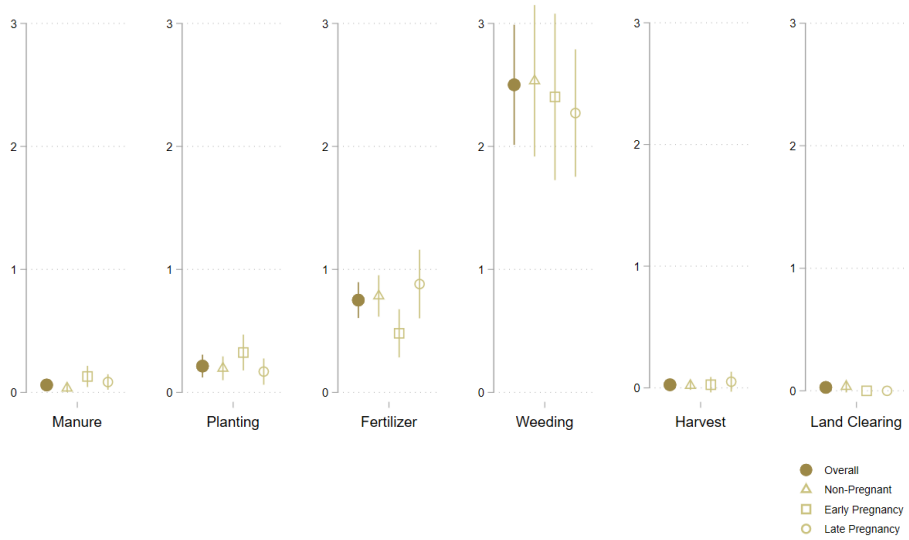
Program Implementation

Of the 440 farmers who were selected to receive the treatment and were confirmed to be eligible, 354 (80.45%) made use of the Planting Partners program (Table A.10). Farmers who opted to participate in the program utilized an average of 13.59 hours of casual laborers' time, with each workday with the casual worker being roughly 3.5 hours. While this is a shorter span of time than what was initially stipulated, feedback from both farmers and casual workers conveyed that six hours was too long to be actively working on the farm. This is supported by the average number of hours reported by farmers in the control group.[12] Participants utilized the program primarily for weeding tasks (2.5 days), and utilization did not vary by pregnancy status (Figure 1 and Table A.1). Generally, beneficiary farmers rated the casual laborer assigned to them as better-skilled than the average farmer.

[11] While marital status and land ownership *are* associated with higher yields in the control group, these margins of imbalance cannot explain our results. If marital status explains our results, then treatment effects should appear similar among women in the early- and late-pregnancy groups, since these groups are both more likely to be married. Instead, treatment effects among women in the non-pregnant group and the late-pregnancy group appear similar, even though imbalance on marital status goes in opposite directions. Similarly, total acres owned is only unbalanced in the late-pregnancy group, where there are no treatment effects.

[12] Non-pregnant farmers reported working four hours per day on average.

Figure 1: Number of days of Planting Partners labor redeemed per activity



Note: Error bars represent 95% confidence intervals. Standard errors calculated via wild cluster bootstrapping.

3 Background: Agriculture and Fertility

3.1 Maternal Malnutrition and Female Farming

Globally, 8% of adolescent girls and 10% of women are considered underweight, with 68% of underweight women and girls living in South Asia and sub-Saharan Africa (UNICEF, 2022). Beyond female malnutrition's direct contribution to the gender gap via the reduction in learning and work potential, maternal malnutrition is associated with increased risks of many adverse health outcomes (de Souza et al., 2011; Talebi et al., 2024; Allen et al., 2025; Lechtig et al., 1978; Dolati et al., 2025).[13] The issue of maternal malnutrition is further exacerbated in settings in which subsistence farming is prevalent. At a time when it might be optimal to conserve their labor due to the physiological constraints of pregnancy symptoms and increased nutritional needs, pregnant women must conduct strenuous physical labor during the agricultural season to improve food security in the harvesting period (Talebi et al., 2024; Khammarnia et al., 2024; Marshall et al., 2022; Rao et al., 2003).

At the same time, female-managed plots have systematically lower productivity relative to male-managed plots. The underdevelopment of female-managed plots both directly limits total household income for households primarily engaged in subsistence farming and

[13]For example: low birthweight, reduced child development, gestational hypertension, gestational diabetes, and infant mortality.

undermines women's ability to influence household decisions, including fertility timing (Bernard et al., 2020; Doss, 2013). Given women's propensity to spend on food and healthcare when they have more financial power, this has implications for the well-being of all household members (Duflo, 2003; Kevane, 2012; Lufuke and Tian, 2024).

Located in southern Africa, Malawi is an ideal setting to explore this tension. Its economy strongly relies on farming, with approximately 80% of households engaged in agricultural activities (Akram-Lodhi, 2018). A majority of agricultural activity is small-scale for the purpose of providing food and income to the household. There is also a gendered dynamic to the types of crops grown by women and men, with women tending to cultivate food crops like maize, soybeans, and pigeon peas, and men tending to cultivate cash crops like tobacco and cotton (Zimba et al., 2025; Perelli et al., 2024). Similar to other countries in this region, there is a 28% gap in productivity between male-managed and female-managed plots (Akram-Lodhi, 2018; UN Women, 2015; Burke and Jayne, 2021).

While women in Malawi are actively engaged in farm labor, they also are very frequently pregnant or postpartum. In a cross-sectional representative survey, forty percent of women in rural Malawi aged 18–35 years are pregnant or have a child aged less than two years, and an additional 11% want to have another child in the next two years (National Statistical Office (NSO) et al., 2024).

3.2 Seasonality of Fertility and Agriculture

3.2.1 Data and Empirical Specification

Using data from both the OAF administrative survey and the endline survey, we can measure the differences in agricultural behavior by pregnancy status to better expand the empirical evidence base for reduced investment into farming due to pregnancy and postpartum. We restrict our sample to data collected in control sites and estimate the differences in land renting; land usage; productivity toward the beginning of the agricultural season measured in hours and lines worked per day; productivity measured during endline when recalling the average number of hours worked per week; engagement in non-agricultural income generating activities; and total harvest yield. Land usage is defined as the proportion of land that is owned or rented by the household that is cultivated during the agricultural season under observation. Similarly, we describe the differences in food security during the post-harvest period by pregnancy status among farmers in control sites.

We use data from the Malawi 2024 Demographic and Health Survey (MDHS 2024) to evaluate how natality trends correlate with the agricultural season. We formally explore whether there is monthly and quarterly seasonality in these trends using the following

two models:

$$\text{NumBirths}_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{Month}_t + \epsilon_t, \tag{1}$$

$$\text{NumBirths}_t = \beta_0 + \beta_1 \text{Time}_t + \beta_2 \text{Quarter}_t + \epsilon_t. \tag{2}$$

We estimate these OLS regressions using time-series data of the total number of birth events per month t in the MDHS 2024 from January 2013 to December 2023, with Time_t being a continuous measure of time in months. Month_t and Quarter_t are categorical variables for the month of the year and the quarter of the year, respectively. We estimate standard errors using the Newey-West estimator, which is robust to temporal autocorrelation.

3.2.2 Descriptive Evidence

Agricultural behavior by pregnancy status

Table 2: Agricultural behavior by pregnancy status

Measure	(1) Non-Pregnant (N=190)	(2) Enrolled 1st/2nd trimester (N=75)	(3) Enrolled 3rd trimester (N=60)	p-value
Acres rented by the household for use	0.35 (0.04)	0.36 (0.07)	0.32 (0.09)	0.882
Proportion of household land cultivated	0.95 (0.01)	0.96 (0.01)	0.96 (0.02)	0.891
Whether the household rents out land	0.08 (0.02)	0.04 (0.02)	0.02 (0.02)	0.109
Reported lines worked per day at season start	22.98 (1.11)	22.79 (1.64)	21.15 (2.01)	0.460
Reported hours worked per day at season start	4.34 (0.13)	3.85 (0.21)	4.25 (0.25)	0.411
Average hours of labor per week from self at season end	22.72 (0.76)	18.84 (1.35)	21.05 (1.53)	0.067
Whether the farmer engages in non-agricultural income generating activities	0.67 (0.03)	0.60 (0.06)	0.58 (0.06)	0.308
Total agricultural yield (USD)	818.23 (60.04)	683.98 (72.02)	707.05 (82.55)	0.212

Note: Results presented as mean (standard error). p-values calculated via simple linear regression of each measure regressed upon pregnancy status binary indicators with district-level fixed effects. Final p-values estimated with wild cluster bootstrap jointly testing indicators against null hypotheses that they equal zero, with clustering at the site-level. Average hours of labor per week recalling over the whole agricultural season winsorized at the 1st and 99th percentiles.

Table 2 presents measures of agricultural engagement by pregnancy status group among farmers in the control group.[14] Land usage does not vary significantly by pregnancy

[14]We assess the statistical significance of differences between groups using OLS models with the agri-

status ($p = 0.891$), with each subgroup cultivating upwards of 95% of available land. If anything, pregnant women rent out *less* land than non-pregnant women ($p = 0.109$), suggesting that women do not allocate their land to renters or fallowing when they are pregnant, and continue to self-cultivate the land. The number of lines and hours worked each day on average toward the start of the agricultural season do not vary ($p = 0.460$ and $p = 0.411$). However, there is a difference in the weekly average number of hours worked when recalling over the entire agricultural season ($p = 0.067$). Farmers enrolled while not pregnant reported 22.72 hours per week, while farmers enrolled in their first/second trimesters and third trimester reported 18.84 hours and 21.05 hours per week respectively. The consequences of this difference in labor supply is evidenced by the qualitatively lower agricultural yields of both pregnancy groups. While the yields measure is noisy and we are unable to statistically reject equality in total yields across pregnancy stages, the magnitude of the gap in yields is economically significant (818.23 USD *vs.* 683.98 USD, a 16% gap).

There are two possible explanations for why our endline measure of productivity is correlated with baseline pregnancy status but our baseline measure of productivity is not. First, the strenuous nature and time-sensitivity of farming activity changes throughout the agricultural season. The endline measure recalls productivity across all of these activities while the baseline measure only references contemporaneous activities, which may be the least sensitive to labor productivity shocks.[15] Second, the endline measure of productivity takes into account the shock of childbirth and the subsequent recovery period.

Food security by pregnancy status

We present differences in food security in Table 3.[16] We find differences across pregnancy stages in the frequency of food consumption. Farmers in both pregnancy groups are more likely to report going to bed hungry a few or more times in the preceding month (35% and 39% *vs.* 27%; $p = 0.137$). Additionally, both pregnancy groups are more than 10 percentage points (pp) less likely to report having at least two meals the day before or two days before ($p = 0.030$ and $p = 0.003$). There is less of a difference in diet quality as

cultural measure regressed on binary indicators for pregnancy status. We include district-level fixed effects and cluster standard errors at the site level. Due to the low number of clusters, p-values for the joint test of the coefficients against the null were derived using wild cluster bootstrapping with 999 replications.

[15]The agriculture season starts with land preparation in August, which is often considered the highest-effort activity; continues with planting in November, which is lower effort; and continues with weeding, which occurs sometime between December and March, and is both high-effort and very time-sensitive. While some women may have been preparing the land at the time of the administrative survey, most were likely engaged in planting, which is both less time-sensitive and less strenuous than weeding. Furthermore, it may be the case that one's pregnancy status does not hinder labor output when timing is flexible – for example, if there is high temporal variation in how well women feel.

[16]Measures of statistical significance are calculated the same way as in Table 2.

Table 3: Food security by pregnancy status

Measure	(1) Non-Pregnant (N=190)	(2) Enrolled 1st/2nd trimester (N=76)	(3) Enrolled 3rd trimester (N=60)	p-value
Whether the farmer went to bed hungry a few or more times in the past month	0.27 (0.03)	0.39 (0.06)	0.35 (0.06)	0.137
Whether farmer had three meals yesterday	0.41 (0.04)	0.29 (0.05)	0.28 (0.06)	0.030
Whether farmer had three meals the day before	0.42 (0.04)	0.26 (0.05)	0.23 (0.06)	0.003
Days of maternal meat consumption in the past month	2.18 (0.21)	2.03 (0.38)	1.95 (0.35)	0.692
Days of maternal egg consumption in the past month	2.56 (0.27)	2.13 (0.41)	2.05 (0.40)	0.479
Days of maternal usipa consumption in the past month	4.11 (0.42)	3.66 (0.63)	3.78 (0.69)	0.720
Days of maternal fish consumption in the past month	3.99 (0.33)	4.51 (0.56)	2.78 (0.46)	0.124

Note: Results presented as mean (standard error). p-values calculated via simple linear regression of each measure regressed upon pregnancy status binary indicators with district-level fixed effects. Final p-values estimated with wild cluster bootstrap jointly testing indicators against null hypotheses that they equal zero, with clustering at the site-level.

proxied by frequency of protein consumption.

Fertility trends in Malawi

These correlations may not point towards a large problem if most women are able to time their pregnancies so that they are not overlapping with the agricultural season.[17] Instead, using nationally representative data, we find that births are *more* likely to happen during the peak of the agricultural season.[18]

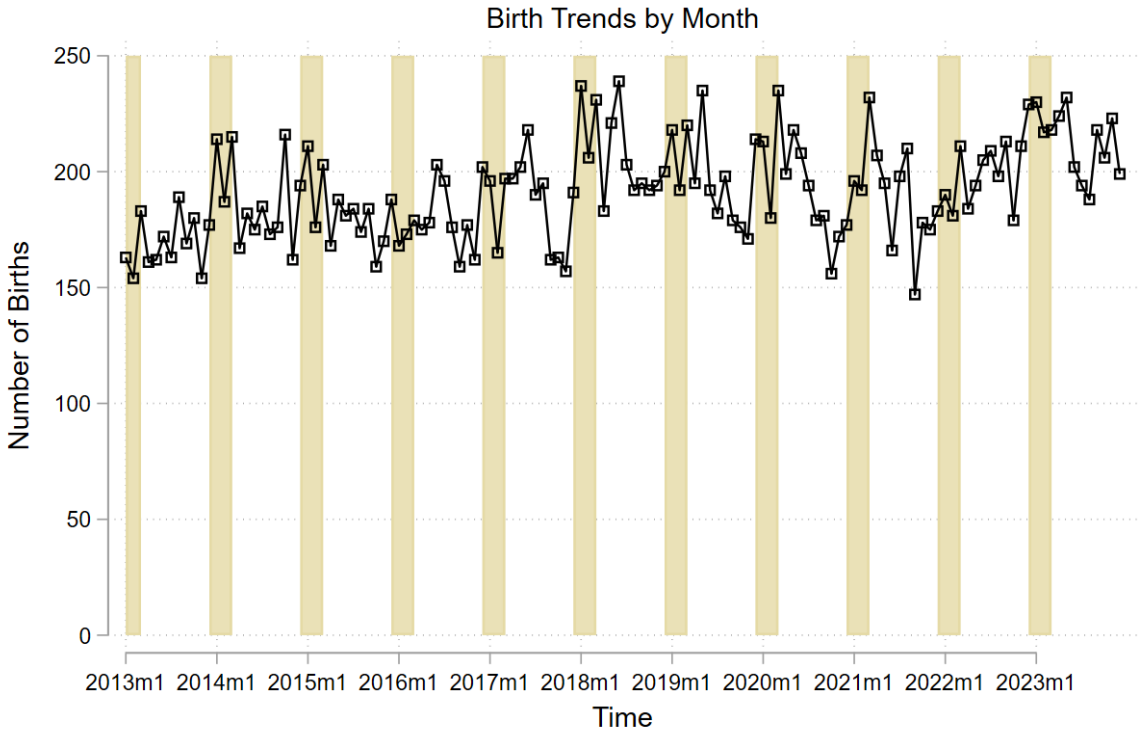
We illustrate the monthly trends in childbirth events from January 2013 through December 2023 in Malawi using the birth history module of the MDHS 2024 in Figure 2. There is evidence of modest seasonality, with birth events at their lowest between September and November and reaching their peak in the first quarter, specifically January and March (Table 4). This peak in natality coincides with the majority of the lean season and

[17] Bryson et al. (2021) show qualitatively that the tradeoff between time investment into childbirth and farming prompts female farmers to attempt to time their pregnancies around the harvest season in order to both overlap with the time of greatest food availability and least intensive labor requirements. Even so, this is not always possible even with accessible family planning services available and full bodily autonomy.

[18] Our results are largely consistent with Dorélien (2016), which finds evidence of modest seasonality in Malawi using births from 1980-2000.

the peak in rural labor demand in December and January when farmers are engaging in planting and weeding activities (De Weerdt et al., 2026; Famine Early Warning Systems Network and USAID, 2024). The hypothesis that these birth trends are influenced by the agricultural calendar is further supported when the data are disaggregated by rurality, with these trends becoming more pronounced in rural areas and disappearing in urban areas (Figures A.1 and A.2, Tables A.2 and A.3). Dorélien (2016) similarly find a dip in births around November across multiple sub-Saharan African countries, which is consistent with lower conception rates during the beginning of year when there is higher labor demand.

Figure 2: Seasonality in birth events in Malawi



Note: Point estimates represent unweighted total number of birth events that occur in each month in Malawi Demographic health Survey 2024. Blue shading indicates lean season (December through March). Data restricted to January 2013 to December 2023.

Under the assumption that households have total agency over birth timing, these trends suggest suboptimal fertility behavior because the third trimester and childbirth overlap with a period in which calories are both scarcer due to limited food supply and in even greater demand as a result of higher workload. This is amplified by the fact that ensuring sufficient calorie intake becomes even more important for the health of both the mother and child during the third trimester and early postpartum.

We are partially able to explain these trends with premature births. Some combination of the malnutrition caused by food scarcity and strenuous labor may influence birth timing

Table 4: Models of seasonality of birth events in Malawi

(1)	
Month	Estimate (SE)
January	28.43*** (7.91)
February	8.82 (5.64)
March	35.93*** (6.98)
April	11.68** (5.42)
May	24.79*** (6.53)
June	20.35** (8.06)
July	14.37** (5.56)
August	11.76* (6.49)
September	3.41 (4.76)
October	3.07 (7.17)
November	REF
December	18.20** (5.94)

(2)	
Quarter	Estimate (SE)
First Quarter	17.31*** (4.61)
Second Quarter	11.85** (4.70)
Third Quarter	2.76 (2.87)
Fourth Quarter	REF

Note: OLS regressions of total number of birth events per month in model (1) and quarter in model (2), with a continuous measure of time in months. Results presented as point estimates (standard error), with Newey-West standard errors allowing for temporal autocorrelation between time periods, with optimized lags (a lag of 9 time periods in model (1) and 11 in model (2)). ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

by inducing women to give birth prematurely (Dole et al., 2003; Bonell et al., 2022). The prevalence of birth events that end in preterm birth is highest in January (11.27%), March (11.45%), and May (11.60%) (Table A.4). If we assume that these premature births were accelerated by roughly a month, then it could be that a greater portion of households were planning the birth and the subsequent postpartum period to overlap with the early and middle stages of the harvest season when food is most available. Given that this would be the point at which the maternal consumption requirements are highest, this would be indicative of a fertility pattern that aims to optimally match the demand and supply of calories (Kominiarek and Rajan, 2016). However, the differences in preterm prevalence rates by month are not large enough to explain away all of the excess bulk of childbirths in the first quarter.

An alternate explanation extrapolates away from planned fertility timing, and instead considers the presence of conducive factors for conception. Assuming an average gestation

period of nine months, births that occur in the first quarter were conceived in the harvest period. The reduced stress from lower labor demand and increased food availability during the harvest period may jointly support higher fertility rates, resulting in more births in the first quarter (Lewinski and Brzozowska, 2023; Gaskins and Chavarro, 2018).

The fact that we see marked differences in productivity and food security in our sample between women who gave birth during the agricultural season and those who did not raises the question whether there are differential unobserved characteristics (i.e. access to resources or agency within the household) that influence some women to 'select' into giving birth during the peak of the labor market. However, this does not seem to be the case when looking at total fertility history of the women in control sites (Table 5). Almost every farmer in each of the three pregnancy groups reported giving birth to at least one child during either the period we categorize as an 'early pregnancy' (February - July) or 'late pregnancy' (December - January). Furthermore, we do not see evidence of learning, whereby a female farmer experiences a childbirth during the agricultural season and then adjusts her future fertility timing. Farmers in each of the pregnancy groups have had between an average of 2.77 (non-pregnant group) to 3.12 (late pregancy group) births during the agricultural season. Thus, we take pregnancy timing within our sample to be a quasi-random labor productivity shock, rather than a correlate of unobserved characteristics.

3.3 Rural Labor Markets in Malawi

Participation in Malawi's rural casual labor markets is prevalent, with roughly 75% of farming households engaging in the supply of short-term labor, known locally as *ganyu* (Benson and De Weerdt, 2023). There is less evidence on the demand for *ganyu*, and the details of what this work entails. Rural labor markets are often characterized by liquidity constraints in which many farmers do not have the funds or credit readily available to purchase short-term labor, even in situations where a small investment in labor would yield better returns during harvest (Ambler et al., 2018; Fink et al., 2020). Furthermore, while there is evidence that *ganyu* are locally sourced from the same village or nearby villages to the employing farm, there is still the possibility for search costs of finding *ganyu* when worker quality is not perfectly observable (Ambler et al., 2018; Benson and De Weerdt, 2023; Breza and Kaur, 2025).

We can again leverage data collected during our study procedures to describe behaviors around demand for *ganyu* in this demographic of smallholder land-owning farmers. We collect data on the number of laborers hired, hours worked, and wages paid across five farming activities: planting (November - December), fertilizer application (November - January), weeding (December - February), harvesting (March - July), and land clear-

Table 5: History of fertility behavior and the agricultural season

	Group	Number of Early Pregnancies	Number of Late Pregnancies	Number of Pregnancies
A.	Non-Pregnant	1.93 (0.10)	0.85 (0.07)	2.77 (0.11)
	Early Pregnancy	2.28 (0.13)	0.81 (0.13)	3.09 (0.18)
	Late Pregancy	1.40 (0.15)	1.72 (0.13)	3.12 (0.21)

	Group	Proportion of births Early Pregnancies	Proportion of births Late Pregnancies	Proportion of births Either
B.	Non-Pregnant	50.22% (2.21)	21.83% (1.74)	72.04% (1.99)
	Early Pregnancy	64.25% (2.98)	19.09% (2.52)	83.35% (2.15)
	Late Pregancy	34.54% (2.81)	49.28% (2.84)	83.82% (2.45)

	Group	Ever had an Early Pregnancy in prior seasons	Ever had a Late Pregnancy in prior seasons	Ever had any Pregnancy in prior seasons
C.	Non-Pregnant	86.84% (2.46)	55.26% (3.62)	95.26% (1.55)
	Early Pregnancy	77.33% (4.87)	53.33% (5.80)	94.67% (2.61)
	Late Pregancy	80.00% (5.21)	45.00% (6.48)	88.33% (4.18)

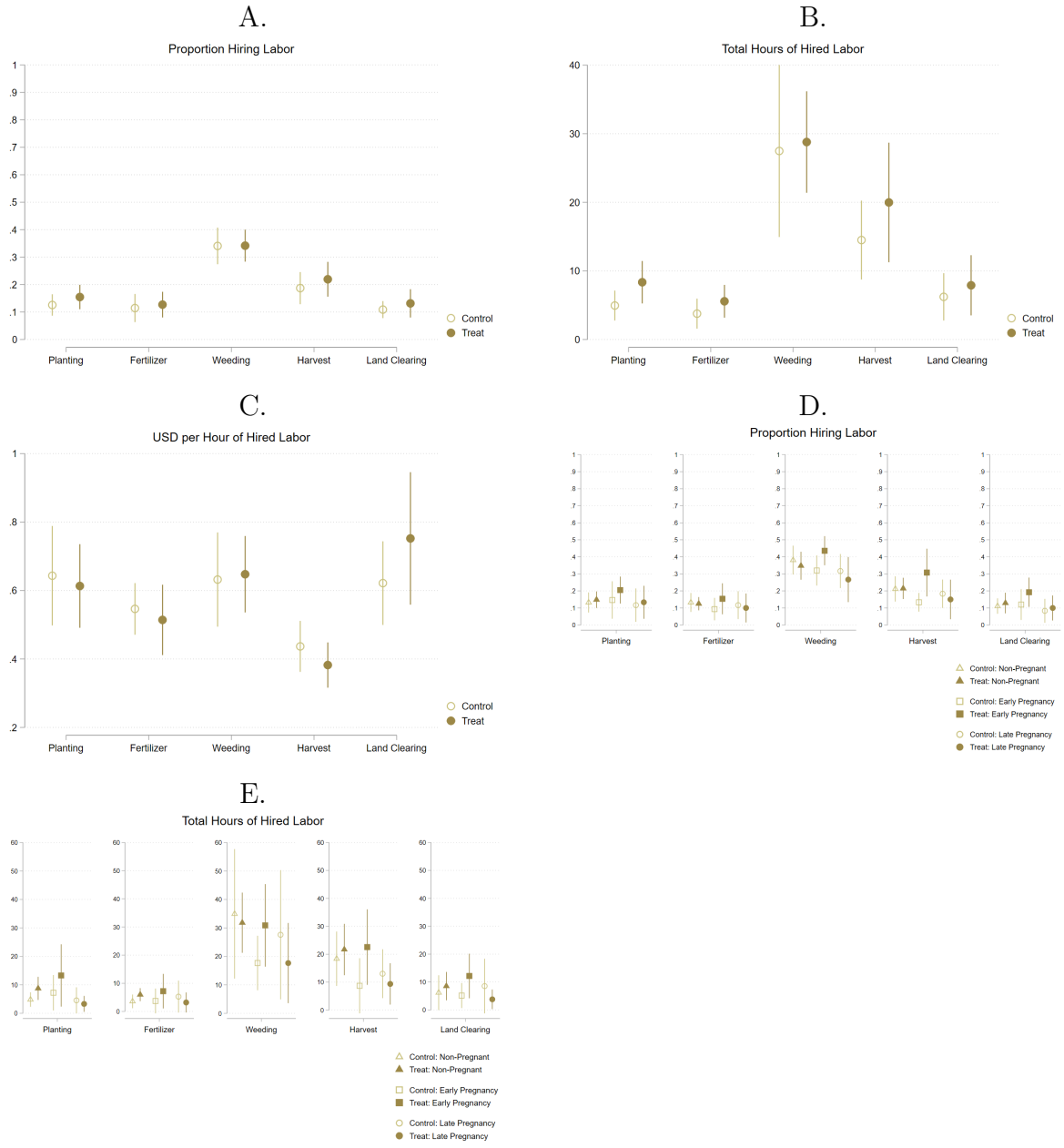
Note: Data limited to those collected in control sites. Panel A shows average number of births that would have been categorized as 'Early Pregnant', 'Late Pregnant', or either. Panel B shows the average proportion of all births that would have been categorized as 'Early Pregnant', 'Late Pregnant', or either. Panel C shows prevalence of having at least one birth in a previous agricultural season that would have been categorized as 'Early Pregnant', 'Late Pregnant', or either.

ing/burning (June - August). In line with prior research on rural labor in this context, demand for labor is highest during the weeding period (Figure 3, which shows raw data on the probability of hiring any *ganyu* stratified by treatment groups and by farming activity). Among farmers in control sites, 34% hired at least one casual worker for weeding activities, while the next highest rate of market engagement was during harvest (19%). Farmers in control sites hired an average of 55 hours of *ganyu* labor across the entire farming season, with approximately half of this labor allocated toward weeding. Average wages for casual labor were relatively stable throughout the agricultural season, though there is a slight dip during harvest activities.[19] Despite the gap in self-sourced labor between pregnant and non-pregnant farmers that we describe in Section 3.2.2, pregnant farmers do not compensate for this deficit with hired labor, either in terms of market

[19]While we cannot say why wages are only lower in the harvest, it is conceivable that hiring farmers have a lower willingness-to-pay for labor, and *ganyu* are willing to take on low-wage labor, given seasonality in liquidity and consumption. Labor hired during the harvest must be paid before the earnings from the harvest are realized, implying that this is a period of intense resource constraints at the tail-end of the lean season.

participation or total number of hours.

Figure 3: Casual labor consumption



Note: Error bars represent 95% confidence intervals. Standard errors calculated via wild cluster bootstrapping.

4 Results: Main Outcomes

4.1 Empirical Specification

4.1.1 Intention-to-treat effects and effect heterogeneity

We estimate the effect of assignment to treatment using the following linear regression model specification:

$$Y_{ids} = \beta_0 + \beta_1 T_s + X_i\beta + \delta_d + \epsilon_{ids}, \tag{3}$$

where T_s indicates whether farmer i is in a treated site, and δ_d are district fixed effects. We include vector X_i containing variables that are unbalanced in either the main sample or within pregnancy groups and whether the participant gave birth during the trial in order to improve efficiency and correct for residual imbalance.[20] Standard errors are clustered at the site level.

We pre-specified our hypothesis that giving birth during the agricultural season would be a greater negative shock to self-sourced labor supply, relative to caring for an infant but not going through pregnancy. As such, we evaluate our pre-specified analysis of the impact of the labor subsidy heterogeneously by pregnancy term at the time of program start by estimating the following model:

$$\begin{aligned} Y_{ids} = \beta_0 + \beta_1 T_s + \beta_2 \mathbb{I}(P_i = 1) + \beta_3 \mathbb{I}(P_i = 2) \\ + \beta_4 \mathbb{I}(P_i = 1) \times T_s + \beta_5 \mathbb{I}(P_i = 2) \times T_s + X_i\beta + \delta_d + \epsilon_{ids}. \end{aligned} \tag{4}$$

Here, P_i is a categorical variable for pregnancy status at the time of recruitment, where 0 indicates non-pregnant (62% of participants), 1 indicates first or second trimester (21% of participants), and 2 indicates third trimester (17% of participants).[21] We pool women who were in the first two trimesters of pregnancy because there are very few women who were in their first trimester of pregnancy during recruitment, likely because they were not aware of or unwilling to disclose their pregnancy status at the time (45 women in trimester 1, compared with 108 women in trimester 2). While X_i are the same covariates as Equation 3, we also allow interactions between P_i and variables that are unbalanced across treatment and control within any of the sub-samples to enter the set of variables

[20] The unbalanced variables are a continuous measure of household size as measured in the administrative data collected before treatment assignment (unbalanced in the pooled sample, and among non-pregnant farmers), a binary indicator for whether the participant is a married and cohabitating with partner (unbalanced in groups individually, but not the pooled sample, since effects go in different directions within sub-groups), a continuous measure of age in years (unbalanced within the early-pregnancy group), and the number of acres owned by the household (unbalanced in the late pregnancy group) (Tables A.6 - A.8)

[21] Because our estimations of pregnancy beginnings are based on dates of child birth, we omit this categorization for women who experienced a miscarriage during the agricultural season. The probability of miscarriage (10%) is unaffected by treatment status.

that a double-selection Lasso algorithm can select, given that some of these variables go in different directions for different sub-groups (Belloni et al., 2014). Data were analyzed using Stata version 18.0 (StataCorp, College Station, TX).

4.1.2 Treatment-on-the-treated effects

Treatment take-up is imperfect, with 20% of treated households never utilizing Planting Partners. Among households who *did* utilize Planting Partners, the hours of Planting Partner time on the farm ranged from 1 to 40 hours (mean = 13.5 hours, median = 12 hours).[22] To estimate the impacts of Planting Partners support on the farm, we estimate treatment-on-the-treated (TOT) effects via two-stage least squares:

$$Y_i = \beta_0 + \beta_1 \hat{Z}_i + X_i\beta + \delta_d + \epsilon_{ids} \tag{5}$$

$$Z_i = \beta_0 + \beta_1 T_s + X_i\beta + \delta_d + \epsilon_{ids}, \tag{6}$$

where Z_i is treatment take-up, and all other variables are as in Equation 3. In our main tables, Z_i is a variable that takes the value of 1 for treated households who used any amount of Planting Partners support on the farm, and takes the value of 0 for control households and treated households who never utilized Planting Partners.[23]

To evaluate TOT effects heterogeneously, we estimate Equations 5 and 6 separately in the three sub-samples among whom we evaluate ITT effect heterogeneity: women who were not pregnant at the time of recruitment, women who were in the first or second trimester at the time of recruitment, and women in the third trimester. We additionally include results in the Appendix estimated using an approach based on Abadie's kappa that is robust in the presence of covariates (Abadie, 2003; Słoczyński et al., 2025).

4.2 Pooled ITT and TOT effects on main outcomes

In this section we show treatment effects on our main outcomes: yields, maternal food security, child solid food consumption, and breastfeeding frequency. Table 6 reports estimated effects of being assigned to treatment. First, there is a notable yet noisy increase in the value of the households' harvest, equivalent to approximately a 10% increase in output when compared to the control farmers. Disaggregating production results by crop, the largest increase is in maize, followed by a relatively large and statistically significant increase in soybean yields (Table A.16). While our estimate of the effect on total yields is noisy, the fact that the increases in crop production are primarily concentrated in the two crops most directly related to the intervention is reassuring.

[22] Planting Partners hours on the farm is winsorized at the first and ninety-ninth percentiles.
[23] There were no cases of Control farmers who were able to access Planting Partners farm labor.

There is a non-significant 0.08 standard deviation (SD) increase in maternal food security, no effect on solid food consumption of the youngest child, and a noisy but negative coefficient on the frequency of breastfeeding. Table A.18 shows TOT effects on main outcomes, which are qualitatively similar.

Table 6: Pooled intention to treat effects on main outcomes

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Total agricultural yield (USD)	70.89 (61.41)	750.67 (38.72)	791	71
Maternal food security	0.08 (0.09)	0.00 (0.05)	791	71
Child solid food consumption	-0.01 (0.05)	0.00 (0.05)	791	71
Average daily frequency of breastfeeding	-0.54 (0.37)	7.67 (0.29)	785	70

Note: OLS models adjusted for whether the farmer gave birth during the trial, total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Maternal food security a standardized composite measure of reports of hunger, meal frequency, and protein consumption. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

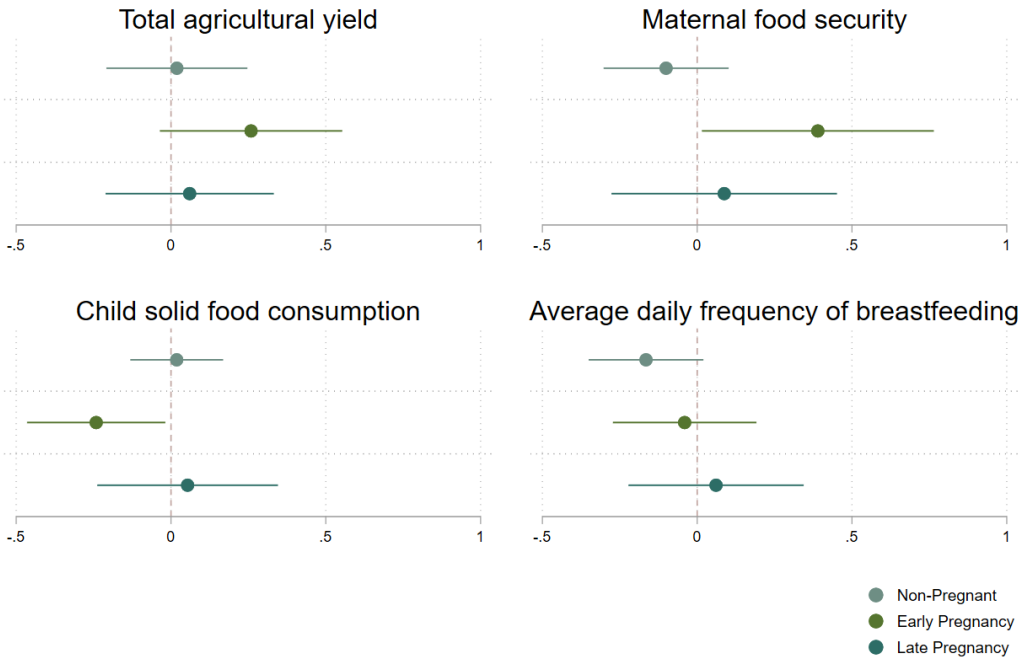
4.3 Effect heterogeneity by pregnancy status at program start

Because we hypothesize that limitations on self-sourced labor would vary across the stages of pregnancy, childbirth, and postpartum, we estimate flexible models to explore whether treatment effects are heterogeneous by pregnancy status at the time of program rollout. Women who were in their first or second trimester at the start of the trial began their third trimester sometime between November and May, overlapping with the start of planting and weeding. They subsequently gave birth between January and July, overlapping with the end of weeding, the harvest, or after the farming season. Those who were in their third trimester all gave birth between November and January. We show results from this analysis in Tables A.20 and A.23.[24] To facilitate the comparison of effect estimates across select outcomes, we illustrate effects on variables standardized by the mean and standard deviation from the control sites in Figure 4.

There is no effect on total yields in the non-pregnant group and a negligible, non-significant change in the late pregnancy group (0.06 SD). In contrast, the effect of treatment assignment on yields is much more pronounced for those in their first or second

[24]We include p-values adjusted for multiple hypothesis testing as described by Romano and Wolf (2016) in Table A.22. The treatment effect on yields in the early-pregnancy group is not significant after correcting for multiple hypothesis testing, which is not surprising given small samples, the conservativeness of Romano and Wolf (2016), and the high variance of the yields outcome. However the interaction between treatment and being in the early pregnancy group has an effect on maternal food security with $p = 0.109$ after adjusting for multiple hypothesis testing, which increases our confidence that this is not a spurious effect. We similarly include q-values calculated in line with Benjamini and Hochberg (1995) in Table A.21, which are again non-significant after adjustment, but have the smallest p-values for all outcomes for treatment effects among women in the early-pregnancy group.

Figure 4: ITT effects on main outcomes by pregnancy group



Note: Point estimates represent the ITT for each subgroup. Maternal food security a standardized composite measure of reports of hunger, meal frequency, and protein consumption. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption. Error bars represent 95% confidence intervals. Outcomes are standardized.

trimester at the start of the agricultural season (0.26 SD, $p < 0.10$). Comparing the treatment effect in dollars gained with the price of the intervention, we find that the benefit of the program is more than ten times the cost: the provision of casual labor at a cost of 18 USD (assuming the redemption of five days of labor) resulted in a 190 USD increase in yield for this subgroup. In line with the prior result, we also find that treatment assignment increased maternal food security by 0.39 SD ($p < 0.05$) among farmers in the early pregnancy group.

Two unexpected findings from this analysis are estimated reductions in child solid food consumption among the early pregnancy group (-0.24 SD, $p < 0.05$) and breastfeeding frequency among the non-pregnant group (-0.17 SD, $p < 0.10$). Both treatment effects are likely artefacts of the feeding natures of the youngest children for both of these groups. For the early pregnancy group, more than 90% of households in treatment and control sites report currently breastfeeding and few of the children born to this group during the trial are old enough to transition to a solid food-based diet (World Health Organization, 2026). Improvements in maternal food consumption directly affect child nutrition via increased quality and/or quantity of breast milk, which is aligned with evidence in public health.[25]

[25]An RCT in Guatemala shows that maternal nutrient supplementation increases exclusive breastfeeding (González-Cossío et al., 1998), and qualitative evidence in several African countries demonstrate

Reductions in breast-feeding frequency among treated women in the non-pregnant group also may indicate improvements in behaviors recommended for child health. Because their youngest children are aged between 1 and 2 years, solid food intake plays a much larger role in ensuring healthy child development (WHO, 2023). While we do not observe changes in the frequency of consumption of solid foods among these children, we cannot rule out that these reductions in breastfeeding are compensated for by other adjustments like increased portion sizes.

We present the TOT effects within each pregnancy group (Figure A.4, Tables A.24-A.26). Among farmers in the early pregnancy group who redeemed any labor from Planting Partners, Planting Partners increased agricultural production by 0.38 SD ($p < 0.10$) and maternal food security by 0.52 SD ($p < 0.05$).

Are these large treatment effects feasible, given the intensity of the treatment? Recall that farmers who made use of the program redeemed an average of 13.5 hours of labor from the program. While this may seem negligible when dividing up across the agricultural season, note that the agricultural season is comprised of a series of 'lumpy' activities that require prompt, high-intensity labor. This is particularly true for fertilizer application and weeding, where delays in labor can cause substantial reductions in yields (OAF, 2016; Mosisa et al., 2022). Gantoli et al. (2013) experimentally show in Benin that variation in weeding timing and intensity can cause yield losses for maize ranging between 38-65%, which is larger than our treatment effect. Moreover, based on data from farmers in control sites, the provision of 13.5 hours equates to over 20% of the total amount of labor a household hires overall, a non-trivial relative increase.

5 Conceptual Framework: Pregnancy and Agricultural Market Failures

Why does a labor subsidy lead to changes in farm productivity and consumption disproportionately for women who spend more of the farming season pregnant? This conceptual model motivates an exploration of labor and land allocations that can arise from market failures, intra-household norms, and their interaction with pregnancy. The model highlights how failure to substitute for women's labor through land and labor markets can exacerbate the economic consequences of pregnancy.

We depart from the canonical agricultural household model in two ways: we introduce pregnancy as a productivity shock to female farmers' labor, and we introduce a threshold of total on-farm labor supply above which farm productivity is increasing. While existing

that a key reason women introduce solid foods before it is recommended is due to perceived low supply of breastmilk (Ter Goon et al., 2020; Semahegn et al., 2014; de Bruyn et al., 2018).

agricultural household models evaluate the impact of labor productivity shocks (Adhvaryu and Nyshadham, 2017; Afridi et al., 2022; Aragón et al., 2021; Dureja and Negi, 2022; Heath et al., 2022; Kochar, 1999, 1995), and non-convexity in the marginal product of on-farm labor (Eswaran and Kotwal, 1986; Feder, 1985; Fink et al., 2020), our paper is the first, to our knowledge, to jointly incorporate both features.

5.1 Setup

A household produces agricultural output using land and labor. Household members choose labor allocations, land use, hired labor, and leisure. They consume all of their income:

$$\max_{z_w, z_o, L_w, \ell_w, L_o, \ell_o, L_h, A_c, A_r} U(c, z_w, z_o), \qquad \text{subject to}$$

$$\begin{aligned}
\text{farm production} \quad & Y = F(A_c, \phi_f(P)L_w + L_o + L_h + L_{PP}), \\
\text{land use allocation} \quad & \bar{A} = A_c + A_r + A_f \\
\text{the budget constraint} \quad & c = pY + w\phi_{of}(P)\ell_w + w\ell_o + rA_r - wL_h, \\
\text{time constraints} \quad & L_w + \ell_w + z_w \leq \bar{T}_w \qquad \text{and} \qquad L_o + \ell_o + z_o \leq \bar{T}_o, \\
\text{subsistence constraint} \quad & F(A_c, L) \geq \underline{F}
\end{aligned}$$

Utility depends on consumption c, women's leisure z_w, and other household members' leisure z_o, where $U(c, z_w, z_o)$ is strictly increasing, continuously differentiable, and concave. Pregnancy exposure during the agricultural season is denoted by P, and modifies women's farm productivity by $\phi_f(P)$ and off-farm productivity by $\phi_{of}(P)$. $\phi_f(P)L_w$ is women's pregnancy-adjusted effective on-farm labor supply, given farm labor hours L_w and the productivity shock given by $\phi_f(P)$. Farm production Y depends on cultivated land A_c, women's effective farm labor supply $\phi_f(P)L_w$, other household member's farm labor supply L_o, hired labor L_h, and labor supplied by the Planting Partners program L_{PP}. For simplicity, we model these sources of labor as (1) additive and (2) as perfect substitutes except when women are pregnant. Households own land $\bar{A}$, which they choose to allocate towards cultivation A_c or renting out A_r, and they let fallow the remaining land A_f. Women can also allocate labor to off-farm activities and earn $w\phi_{of}(P)\ell_w$, where ℓ_w is women's off-farm labor and w is the prevailing wage.[26] Households face a subsistence constraint, and will always allocate enough labor and land to the farm to produce subsistence consumption $\underline{F}$.

Household consumption c is determined by the budget constraint, where p is the per-unit price of farm yields, ℓ_o is other household members' off-farm labor, and r is the return on

[26] For simplicity, we assume that off-farm wages that women and other household members can earn is the same, and that this wage is equal to the wages paid to hired farm labor.

rented-out land. Planting Partners labor L_{PP}, which is provided to households for free, does not enter the budget constraint. The household faces time constraints $\bar{T}_w$ and $\bar{T}_o$.

Assumption 5.1. *The productivity shock of pregnancy makes it profitable to substitute for women's farm labor.*

(a) ***Productivity:*** *Pregnancy reduces women's farm productivity. Pregnancy may also affect off-farm productivity to a lesser extent:*

$$\phi'_f(P) < 0, \qquad \phi'_{of}(P) \leq 0, \qquad |\phi'_f(P)| > |\phi'_{of}(P)|.$$

(b) ***Profitability of Farm Production Relative to Outside Options****: There exists a range of total effective on-farm labor* $\underline{L} < L = \phi_f(P)L_w + L_o + L_h < \bar{L}$ *where farm labor is more profitable than fallowing and working off-farm. We assume that* $F(A_c, L)$ *is continuously differentiable, increasing, and concave in both arguments on* $(\underline{L}, \bar{L})$*:*

$$p \cdot \frac{\partial F(A_c, L)}{\partial L}\bigg|_{\underline{L}<L<\bar{L}} > w$$

(c) ***Profitability of Renting Out Relative to Farm Production****: There exists a high level of pregnancy exposure* $\bar{P}$ *such that it is more profitable to rent out land than cultivate land without help:*

$$r > p \cdot \frac{\partial F(A_c, L)}{\partial A_c}\bigg|_{L=\phi_f(\bar{P})L_w}$$

Together, these assumptions imply that, at sufficiently high pregnancy exposure, women's own farm labor is not the optimal marginal input. It is strictly more profitable for the household to substitute through hired labor, other household labor, or land rental, rather than for pregnant women to continue cultivating.[27]

Assumption 5.2. *Low levels of labor are unproductive, but can still produce subsistence.*

(a) ***Profitability of Farm Production Relative to Outside Options when Labor Inputs are Constrained****: Below* $\underline{L}$*, labor is not supplied in high enough quantities to use complementary inputs efficiently (fertilizers cannot be useful unless they are applied with labor, for example), and fallowing is more profitable than*

[27] We focus on household profits for simplicity, but note that pregnancy likely also increases the marginal utility of 'leisure' time through increased returns to activities such as rest and doctor's visits. If we allow the marginal utility of leisure to depend positively on pregnancy, then the optimal point of substituting out women's farm labor happens earlier, weakening how strong Assumption 5.1 has to be.

farm labor:

$$p \cdot \frac{\partial F(A_c, L)}{\partial L}\bigg|_{L \leq \underline{L}} < w$$

(b) ***Labor Requirements to Produce Subsistence****: Given that the choice of cultivated land is greater than a minimal amount required, $A_c^* > \underline{A_c}$, for some level of total effective on-farm labor $L^*(A_c^*) < \underline{L}$, $F(A_c^*, L^*(A_c^*)) = \underline{F}$. That is, even unproductive labor can still produce subsistence.*

5.2 Incomplete Markets and Pregnancy

Given Assumption 5.1, with complete markets, women should shift time into leisure or less-physically-exacting off-farm work when they are pregnant during the farming season. With frictionless markets, households can adjust to a reduction in women's farm productivity by: (i) hiring labor, (ii) reallocating farm and off-farm labor within the household, or (iii) renting out land. If any of these markets function properly, households can rely on that market to offset women's reduced farm labor supply. However, if all of these markets are simultaneously incomplete, then pregnancy reduces effective labor on cultivated land and limits the household's ability to reallocate labor efficiently across activities.

We introduce three constraints that are the result of 'incomplete markets' for land and labor reallocation. For consistency in terminology, we refer to the inability to substitute farm labor between household members as an incomplete market for intra-household labor substitution, even if it is driven by norms:[28]

$$L_h \leq \bar{L}_h, \qquad A_r \leq \bar{A}_r, \qquad \text{and} \qquad L_o \leq \bar{L}_o^N$$

Households face constraints on access to hired labor $\bar{L}_h$, constraints on access to land renters $\bar{A}_r$, and limits to intra-household labor substitution due to norms or bargaining denoted by $\bar{L}_o^N$. We do not take a stance on where these three market imperfections can arise from; for example, constraints on hired labor $\bar{L}_h$ can arise from labor rationing and true quantity shortages, or *effective* quantity shortages due to search frictions or incomplete credit markets.[29] Furthermore, because of Assumption 5.2(a), it is possible

[28] While other work such as Afridi et al. (2022) look specifically at gendered responses to household labor productivity shocks, it is notable that the labor supply shock itself that we look at is *definitionally* gendered. Furthermore, unlike illness or weather, the time and extent of the labor productivity shock of pregnancy are predictable, implying that substituting women's labor supply should be easy to achieve in the absence of frictions (Fink et al., 2020; Rosenzweig and Udry, 2014). Given this context, we model the gender norms that prevent substitution between male and female household labor as a market friction on par with incomplete credit, land, and hired labor markets.

[29] While labor supply may be readily available, credit market frictions and liquidity constraints can make it impossible for a farmer to pay more than $w\bar{L}_h$ for hired labor, limiting the effective labor available

that the labor constraints bind even if households utilize less labor than $\bar{L}_h$ and $\bar{L}_o^N$.[30]

5.2.1 Predictions with Market Frictions

Note that households should never choose total farm labor supply between $L^*(A_c^*)$ and $\underline{L}$ — in this range, households have achieved subsistence production, but could earn more on the margin in off-farm wages. Thus, households should always choose a low level of total on-farm labor $L^*(A_c^*)$ (just enough for subsistence) or high levels of labor (above $\underline{L}$, where labor complementarities imply efficiencies), but not an intermediate level. This non-convexity generates key predictions for labor allocation decisions in the face of a labor productivity shock.

Figure 5 demonstrates a stylized depiction of household labor allocation decisions, with and without the Planting Partners labor supply shock, across varying conditions.[31] Panel (a) represents a case with frictionless markets, where the Planting Partners labor supply shock has no effect on total labor supply on the farm, but it does crowd-out labor that would have been supplied otherwise. When households are severely constrained in their own ability to supply labor and face market constraints on hiring labor and normative constraints on intra-household labor substitution (Panel (b)), the Planting Partners labor supply shock is not sufficient to surpass $\underline{L}$, the point where the marginal productive value of farm labor surpasses the prevailing wage.[32] With or without Planting Partners, these households will only supply enough labor to the farm to meet subsistence $L^*(A_c^*)$. Similar to the complete markets condition, Planting Partners will substitute for hired or household farm labor. Conversely, when households are only modestly constrained in their own ability to supply labor (Panel (c)), the Planting Partners labor supply shock is sufficient to surpass $\underline{L}$. Now, it is profitable to utilize all labor available. If there is slack in either $\bar{L}_h$ or $\bar{L}_o^N$, households pushed above $\underline{L}$ by the Planting Partners intervention will *increase* hired or household member labor, which is now more profitable than off-farm wages.

Table B.1 summarizes tests for market frictions that are motivated by the model (Appendix Section B describes these tests in detail).

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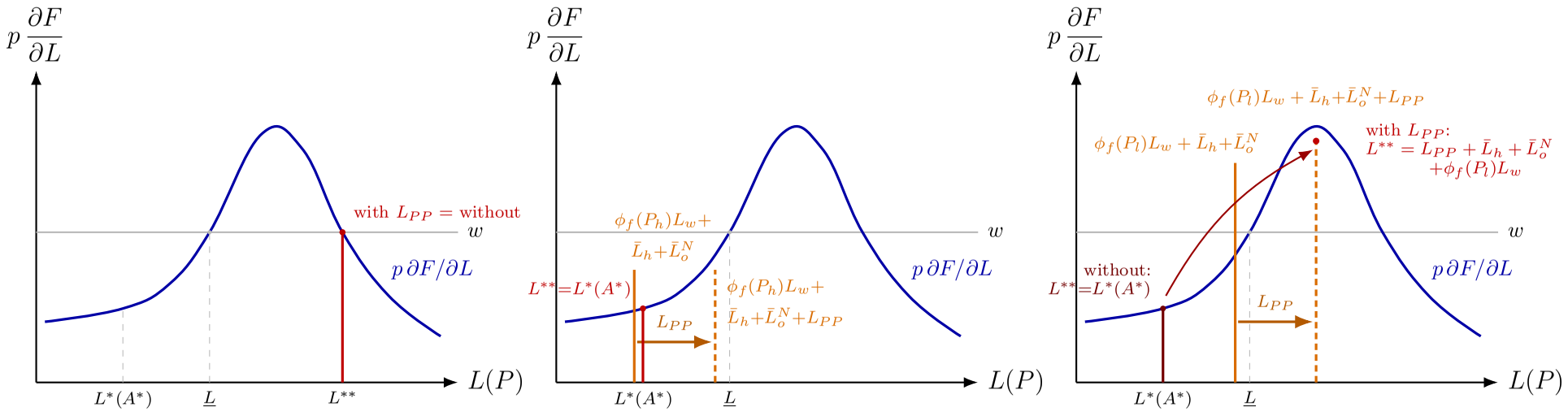
to $\bar{L}_h$.

[30] For example, if a household is $\bar{L}_h + \epsilon$ below $\underline{L}$, she will choose to purchase no hired labor, even if she is able to purchase up to $\bar{L}_h$.

[31] The upper panels of Figure 5 plot household effective on-farm labor against the marginal value of on-farm labor. Figure B.2 in Appendix Section B plots household effective on-farm labor against farm productivity $F(A_c, L)$.

[32] Appendix Figure B.1 visually demonstrates the relationship between pregnancy exposure, the Planting Partners interventions, and total effective on-farm labor.

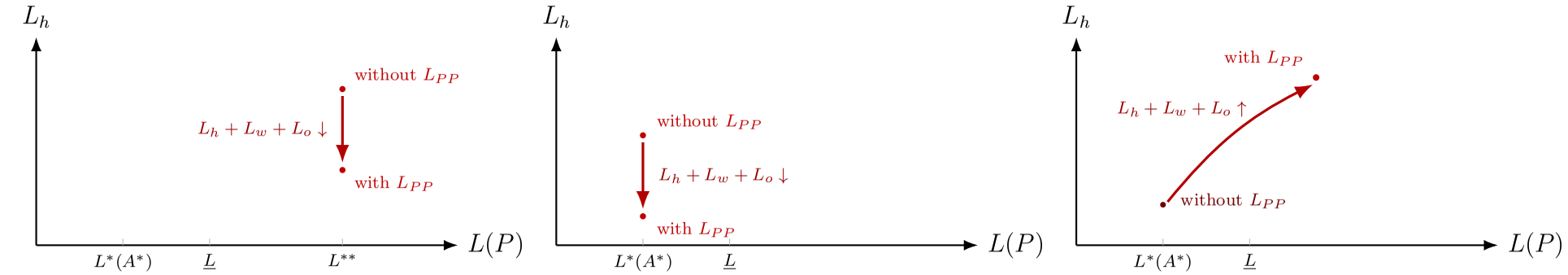
Figure 5: Labor allocations under complete and incomplete markets



(a) Complete markets

(b) Incomplete, $\phi_f(P_{high})L_w + \bar{L}_h + \bar{L}_o^N + L_{PP} < \underline{L}$

(c) Incomplete, $\phi_f(P_{low})L_w + \bar{L}_h + \bar{L}_o^N + L_{PP} > \underline{L}$



(a) Complete markets: L_h falls (displacement with high productivity)

(b) Incomplete markets: L_h falls (displacement with low productivity)

(c) Incomplete markets: L_h rises

Notes: We represent pregnancy and women's own farm labor $\phi_f(P)L_w$ through where the household falls along the horizontal axis (the location of the labor constraint before L_{PP}). Higher pregnancy exposure corresponds to a constraint located further left. The curve $p\,\partial F/\partial L$ is the marginal value product of effective on-farm labor and w is the common wage. $\underline{L}$ is the threshold above which farming becomes more profitable than off-farm work. $L^*(A^*)$ is the minimal about of labor required to meet the subsistence constraint. Columns compare complete markets (a) with incomplete markets (b, c); the L_{PP} shift is identical across (b) and (c). The bottom row plots hired labor L_h, whose response to L_{PP} flips sign from negative (a, b) to positive (c) at the threshold crossing. Parameters are illustrative.

6 Results: Labor and Land Reallocation

In this section we discuss the treatment effects of Planting Partners on households' allocation of labor and land resources by pregnancy group to better contextualize our results and assess the evidence for our model interpretation.

6.1 Labor Allocations

We show the effects of treatment assignment on the number of hours of farm work supplied by hired labor, unpaid farming labor from outside the household, other household members, and the farmer herself. We also report treatment effects on whether the farmer engages in income-generating activities unrelated to agriculture such as piece work or businesses (Figure 6 and Table A.30).[33] One concern with this intervention was that the subsidization of labor for a female-coded crop like maize would cause other household members, particularly male members, to withdraw their support. However, we do not find meaningful reductions in the provision of labor from other household members across all three pregnancy groups, nor any increase in the cultivation of male-coded crops, such as tobacco (Table A.16).

Among non-pregnant farmers, there is a noisy 8 hour reduction in the consumption of hired labor. Though not statistically significant, this may indicate that farmers in this subgroup were not as labor constrained during the agricultural season and were able to substitute the labor received from the program for some of the labor they would have hired otherwise. While it is not clear where this money was reinvested, it also helps clarify why there are no effects on yields within this group. In contrast, we observe that access to Planting Partners led to a noisy 19 hour increase in hired labor for those in the early pregnancy group. This effect is driven by a 14 percentage point increase in the probability of hiring labor during the harvest, which is after the vast majority of women have already utilized Planting Partners ($p < 0.10$, Appendix Table A.31).

This finding points toward a potential channel through which this program impacts yields, and subsequently food consumption. First, pregnancy creates a greater deficit in labor supply or productivity than infant-caregiving, while market frictions and household norms prevent the optimal replacement of the loss in self-sourced labor with other's labor. However, when Planting Partners *ganyu* is utilized for weeding, yields increase, generating labor demand in the harvest season. This is consistent with a model of labor-intensive activities throughout the agricultural season where productivity resembles a Stone-Geary function in which there is a minimum level of labor inputs needed in order for additional labor units to be most productive (Beattie and Aradhyula, 2015). As such, the labor

[33] All measures of labor hours refer to the entire agricultural season and are winsorized at the 1st and 99th percentiles.

provided through this intervention may help farmers achieve this minimum level of labor needed so that hiring additional labor is profitable. There is suggestive support for this hypothesis given that, among the early pregnancy group, there is noisily more labor hired on both the extensive and intensive margins in the treated group during the weeding period (Figure 3D and Figure 3E).

Consistent with this hypothesis, the treatment leads to a large reduction in the hours of hired labor (35 hours, $p < 0.10$) among farmers in the late pregnancy group, many of whom gave birth before or during the weeding period. The magnitude of this reduction is very similar to the total 30 hour 'face value' of the vouchers. Moreover, we find that treatment leads to a 19 pp ($p < 0.05$) increase in the likelihood of engaging in non-agricultural income-generating activities. These two findings in conjunction suggest that, given the extreme physical limitations to engaging in agricultural labor imposed by childbirth and postpartum recovery, farmers in this group reinvest the savings on labor due to this program toward less physically strenuous employment.

Note that, in our model and in these results, the differential patterns across women in early and late stages of pregnancy can be explained by true labor productivity shock differences or by expectations. When women know that they will go through late-term pregnancy and childbirth during the agricultural season, they may feel certain that their labor supply will be severely constrained and plan for less cultivation, leaving little room for Planting Partners to be effective. Conversely, when women enter the farming season early in pregnancy, there may be more uncertainty around how physiologically challenging their pregnancy will be, and they may cultivate land with the expectation of a relatively easy second trimester. This leaves room for Planting Partners to be effective among women who instead have a difficult second trimester. While both mechanisms are consistent with our model and results, the underlying mechanism points to a different policy to attempt to aid women late in pregnancy: it is possible that they simply need more on-farm support (a more intensive treatment), or they may need more advance notice so that they can plan for more on-farm labor than they would expect. While we cannot rule out that expectations play an important role, we do not find that the proportion of land cultivated differs systematically by pregnancy stage in the control group, which is more consistent with a model where expectations do not affect initial decisions.

We do not find reductions in self-sourced labor in any subgroup. This is likely a product of one the stipulations of Planting Partners in which the recipient farmer was required to be present on their farm in a supervisory role when the OAF-provided *ganyu* was there.

Figure 6: ITT effects on labor allocations by pregnancy group



Note: Point estimates represent the ITT for each subgroup. Error bars represent 95% confidence intervals. Outcomes are standardized. Abbreviations: IGA = income-generating activity

6.2 Land Allocations

We show intention-to-treat effects on land allocations in Figure 7 and Table A.33. There are no treatment effects on the likelihood of renting out land in any pregnancy subgroup, suggesting that households are unable to utilize land rental markets to earn income from their land when their labor supply is constrained. The narrative established in Section 6.1, in which we generally find that being offered a labor subsidy induces farmers in the early pregnancy group to invest more into farming while it conversely causes farmers in the late pregnancy group to invest less into farming, extends to household decisions around land allocations. For the early pregnancy group, eligibility for the program leads to a 13 pp increase in the likelihood of renting land for household use and a 0.21 acre ($p < 0.05$) increase in the amount of land rented. On the other hand, farmers in the late pregnancy group cultivate 5 pp ($p < 0.05$) less of the land available to them, without renting out the non-cultivated land.[34]

[34]These patterns suggest there may be an asymmetry in the land rental market in this setting, where it is easier to find households to rent land from than it is to find household to rent land to. Qualitatively, households describe land rental markets as functioning very informally, where individuals seeking more land to rent in simply 'ask around' and households rarely rent land out unless a potential renter approaches them directly. Twenty seven percent of the sample said someone approached them asking if they had land available to rent out, while only 5% said they searched for somebody to rent their land out to, suggesting search frictions may be an important constraint on renting land out.

Figure 7: ITT effects on land allocations by pregnancy group



Note: Point estimates represent the ITT for each subgroup. Error bars represent 95% confidence intervals. Outcomes are standardized.

7 Conclusion

In this paper, we conduct a cluster-randomized trial in order to evaluate a novel intervention to alleviate the labor constraints faced by female farmers where subsistence farming is major source of household income. We find that a direct labor subsidy to women with young children or who are pregnant is most effective in increasing agricultural production for those who spend a majority of the agricultural season pregnant, but who do not go through childbirth until late in the agricultural season or after. Moreover, at least some of the returns to household income from crop production is reallocated toward maternal food consumption. For those enrolled in their first or second trimester, the program alleviates some of the liquidity or quantity constraints that prevent the replacement of lost self-sourced labor, which then encourages re-investment in farming via hiring more labor and renting more land. Conversely, we find that Planting Partners induces women who entered the agricultural season in their third trimester to invest *less* in hired labor and land, and instead reallocate resources to non-agricultural activities. This differential effect between the two pregnancy groups highlights that there is an interaction between the depth of the labor productivity shock and agricultural market failures.

A secondary contribution of this study is that we quantify the magnitude of the shocks to farmer productivity and nutrition that are associated with pregnancy status. Descrip-

tively, among farmers in control sites, when recalling over the entire agricultural season, those who were enrolled into the program earlier in their pregnancy report working fewer hours per week. This is not adequately compensated for by increases in labor inputs from other household members, and there remains a persistent difference in total agricultural production. It is then not surprising that we observe deficits in maternal food security.

One must keep in mind the setting of this study when interpreting these results in order to make inference on how effective the program could be in other settings. Firstly, all of the participants in this trial are members of OAF. Because of the nature of OAF as a non-profit that supports farmers with farming inputs to better promote food security, these member farmers and their families are more likely to be trained as skilled farmers. The vast majority of our sample use precision agricultural practices, and thus the marginal constraint on farm productivity is the physical capacity to supply labor. Secondly, OAF is a well-known and credible third party who farmers trusted to facilitate the selection and administration of a direct labor subsidy.

While our setting is precisely one where the energy to supply labor is the binding constraint and it is feasible to set up a market for a suitable labor substitute, it is also important to note that our comparison across sub-groups pushes us in the opposite direction toward a conservative estimate. The non-pregnant group was still selected into study procedures because they had at least one child aged 1 year or younger. If we assume that domestic labor demands are decreasing as the youngest child ages, then our estimates of the differences in productivity based on pregnancy status are likely lower bounds relative to the average non-pregnant female farmer as the comparison group.

These findings build on prior work demonstrating that supporting women's income-generating activities can improve productivity. However, we show that the fertility outcomes can impact non-domestic labor productivity even before the child is born. Thus, there is a need to develop broader programming that targets pregnant women in addition to women with young children. This is even more important because women whose labor and production is constrained by pregnancy are the very same people where ensuring sufficient nutritional intake is most important for both their own and their children's health and development.

Pregnant women—which, at some point, represents almost all women in this setting—experience significantly greater consequences of the market failures that are characteristic of agricultural economies in developing countries. This finding opens up several promising avenues for future research, including an exploration of the magnitude of "child penalties" for female laborers in developing countries, the consequences of pregnancy and childbirth on the long-term growth trajectories of female-managed plots, and their macroeconomic implications.

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Online Supplementary Material

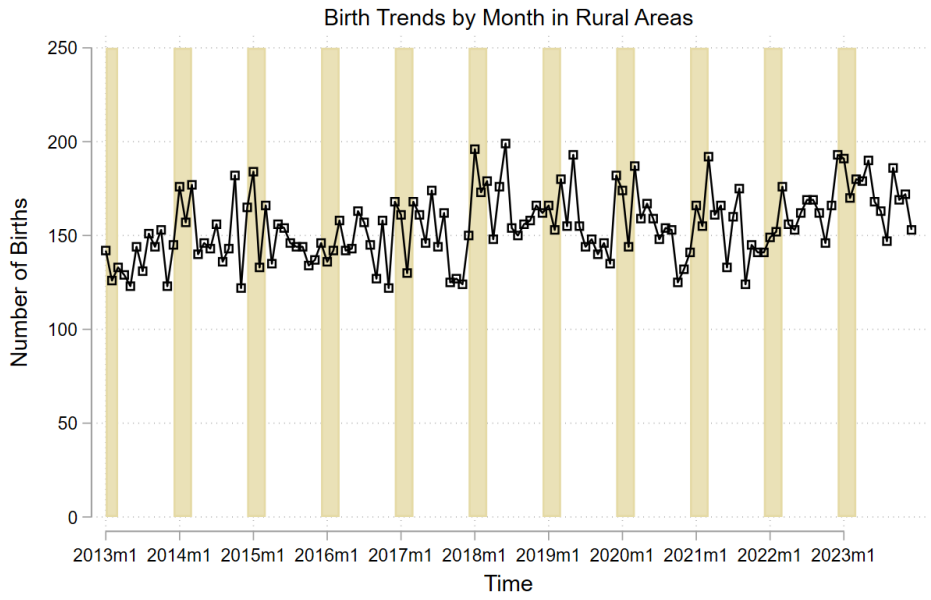
Appendix A Supplementary results

Table A.1: Number of days of Planting Partners labor redeemed per activity

	Manure	Planting	Fertilizer	Weeding	Harvest	Land Clearing
Overall	0.06	0.21	0.75	2.50	0.03	0.03
	(0.02)	(0.05)	(0.07)	(0.25)	(0.02)	(0.02)
Non-Pregnant	0.04	0.20	0.78	2.53	0.02	0.04
	(0.02)	(0.05)	(0.09)	(0.31)	(0.02)	(0.02)
Early Pregnacy	0.13	0.32	0.48	2.40	0.03	0.00
	(0.04)	(0.07)	(0.10)	(0.35)	(0.03)	(0.00)
Late Pregnacy	0.08	0.17	0.88	2.27	0.05	0.00
	(0.03)	(0.05)	(0.14)	(0.26)	(0.04)	(0.00)

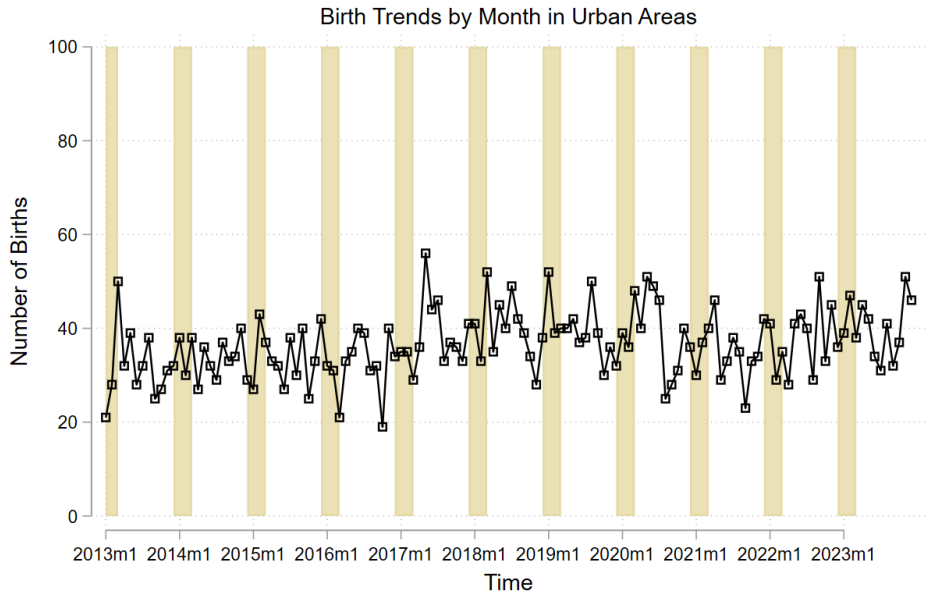
Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

Figure A.1: Seasonality in birth events in rural Malawi



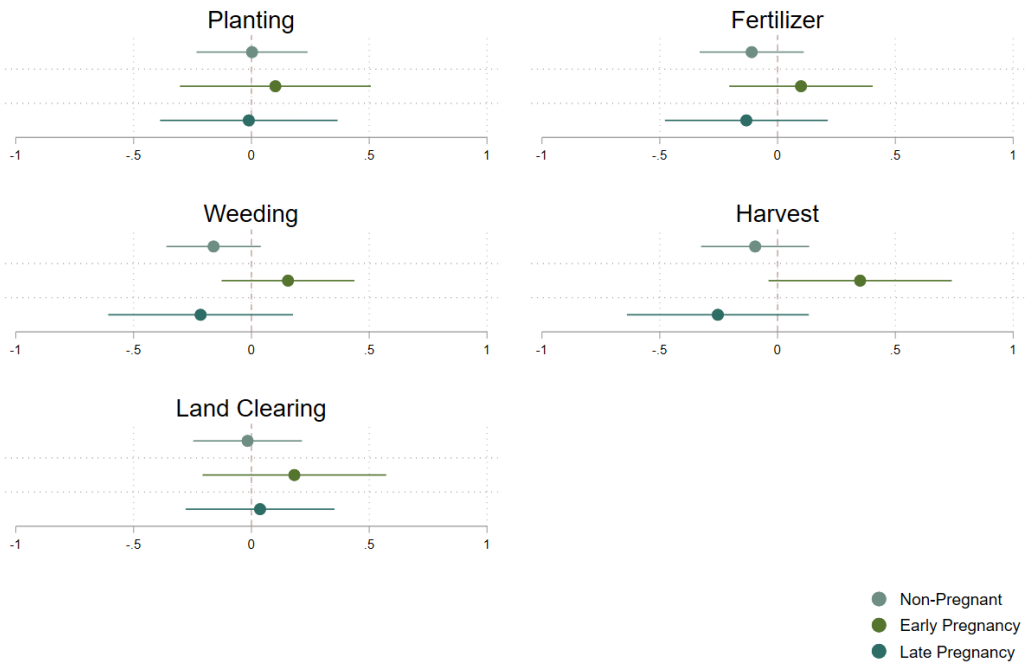
Note: Point estimates represent unweighted total number of birth events that occur in each month in Malawi Demographic health Survey 2024. Yellow shading indicates lean season (December through March). Data restricted to January 2013 to December 2023.

Figure A.2: Seasonality in birth events in urban Malawi



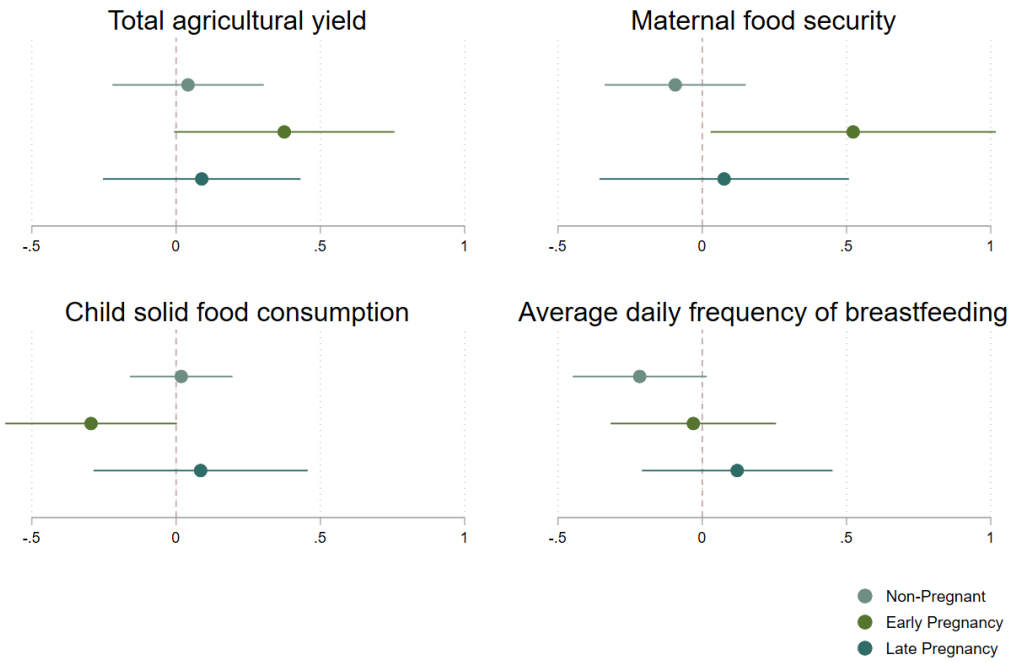
Note: Point estimates represent unweighted total number of birth events that occur in each month in Malawi Demographic health Survey 2024. Yellow shading indicates lean season (December through March). Data restricted to January 2013 to December 2023.

Figure A.3: ITT effects on likelihood of hiring labor for each activity by pregnancy group



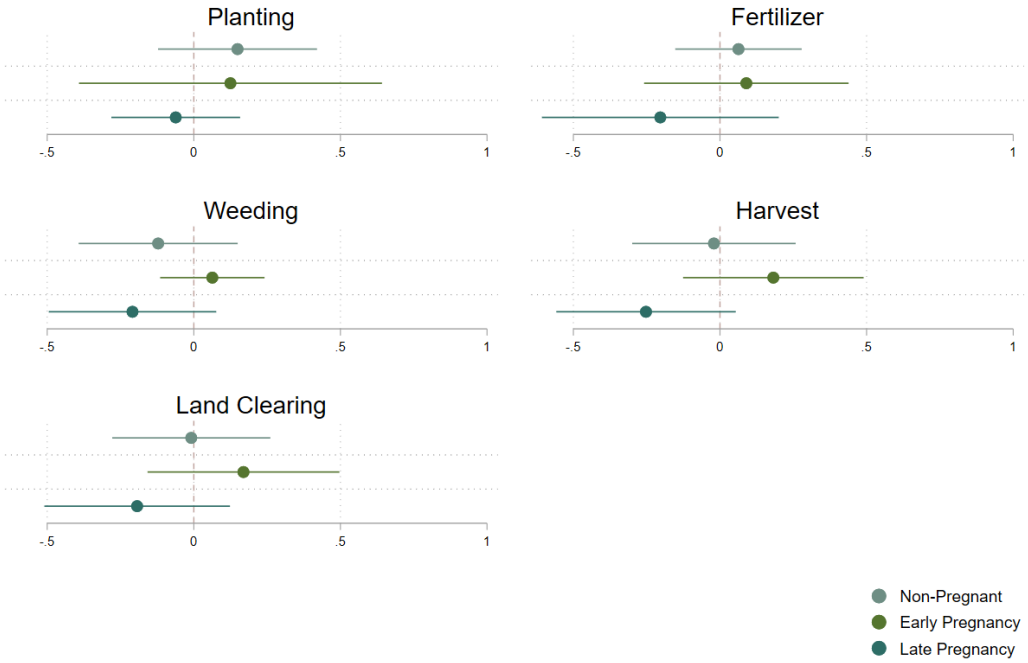
Note: Point estimates represent the ITT for each subgroup. Error bars represent 95% confidence intervals. Outcomes are standardized.

Figure A.4: TOT effects on main outcomes by pregnancy group



Note: Point estimates represent the TOT for each subgroup. Maternal food security a standardized composite measure of reports of hunger, meal frequency, and protein consumption. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption. Error bars represent 95% confidence intervals. Outcomes are standardized.

Figure A.5: ITT effects on total hours of hired labor for each activity by pregnancy group



Note: Point estimates represent the ITT for each subgroup. Error bars represent 95% confidence intervals. Outcomes are standardized.

Table A.2: Models of seasonality of birth events in rural Malawi

(1)	
Month	Estimate (SE)
January	29.31*** (6.34)
February	10.39** (4.08)
March	33.92*** (5.58)
April	12.73** (4.16)
May	21.08*** (5.10)
June	20.43** (7.33)
July	12.78** (4.76)
August	13.40** (6.00)
September	6.21 (4.07)
October	9.56 (7.39)
November	REF
December	18.53** (5.52)

(2)	
Quarter	Estimate (SE)
First Quarter	15.19*** (3.55)
Second Quarter	8.72** (4.12)
Third Quarter	1.44 (2.77)
Fourth Quarter	REF

Note: OLS regressions of total number of birth events per month in model (1) and quarter in model (2), with a continuous measure of time in months. Results presented as point estimates (standard error), with Newey-West standard errors allowing for temporal autocorrelation between time periods, with optimized lags (a lag of 9 time periods in model (1) and 11 in model (2)). ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.3: Models of seasonality of birth events in urban Malawi

(1)	
Month	Estimate (SE)
January	-0.88 (3.41)
February	-1.57 (1.76)
March	2.00 (3.86)
April	-1.05 (2.15)
May	3.71 (3.09)
June	-0.08 (2.74)
July	1.59 (3.34)
August	-1.65 (3.24)
September	-2.79 (2.85)
October	-6.49** (2.30)
November	REF
December	-0.33 (2.14)

(2)	
Quarter	Estimate (SE)
First Quarter	2.13 (1.54)
Second Quarter	3.13* (1.72)
Third Quarter	1.32 (1.68)
Fourth Quarter	REF

Note: OLS regressions of total number of birth events per month in model (1) and quarter in model (2), with a continuous measure of time in months. Results presented as point estimates (standard error), with Newey-West standard errors allowing for temporal autocorrelation between time periods, with optimized lags (a lag of 9 time periods in model (1) and 11 in model (2)). ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.4: Prevalence of preterm births per month in Malawi

Month	Estimate (SE)
January	11.27% (0.71)
February	10.87% (0.72)
March	11.45% (0.69)
April	10.73% (0.71)
May	11.60% (0.72)
June	10.92% (0.70)
July	10.68% (0.70)
August	10.18% (0.70)
September	10.19% (0.70)
October	9.89% (0.69)
November	10.61% (0.71)
December	9.56% (0.66)

Note: Results presented as prevalence (standard error). Standard errors adjusted for clustering at the maternal-level.

Table A.5: Casual labor consumption

A. Proportion hiring labor

	Control	Treat
Planting	0.13	0.15
	(0.02)	(0.02)
Fertilizer	0.11	0.13
	(0.02)	(0.02)
Weeding	0.34	0.34
	(0.03)	(0.02)
Harvest	0.19	0.22
	(0.03)	(0.03)
Land Clearing	0.11	0.13
	(0.02)	(0.03)

B. Total hours of hired labor

	Control	Treat
Planting	4.95	8.35
	(1.11)	(2.01)
Fertilizer	3.77	5.57
	(1.02)	(0.98)
Weeding	27.48	28.79
	(6.86)	(4.53)
Harvest	14.50	19.98
	(3.55)	(3.80)
Land Clearing	6.22	7.90
	(1.62)	(2.24)

C. USD per hour of hired labor

	Control	Treat
Planting	0.64	0.61
	(0.07)	(0.06)
Fertilizer	0.55	0.51
	(0.04)	(0.05)
Weeding	0.63	0.65
	(0.07)	(0.06)
Harvest	0.44	0.38
	(0.04)	(0.03)
Land Clearing	0.62	0.75
	(0.06)	(0.10)

D. Proportion hiring labor

	Planting	Fertilizer	Weeding	Harvest	Land Clearing
Control: Non-Pregnant	0.13	0.13	0.38	0.21	0.11
	(0.03)	(0.03)	(0.04)	(0.04)	(0.02)
Control: Early Pregnacy	0.15	0.09	0.32	0.13	0.12
	(0.06)	(0.03)	(0.05)	(0.03)	(0.05)
Control: Late Pregnancy	0.12	0.12	0.32	0.18	0.08
	(0.05)	(0.04)	(0.05)	(0.04)	(0.04)
Treat: Non-Pregnant	0.15	0.13	0.35	0.21	0.13
	(0.02)	(0.02)	(0.04)	(0.03)	(0.03)
Treat: Early Pregnacy	0.21	0.15	0.44	0.31	0.19
	(0.04)	(0.05)	(0.04)	(0.07)	(0.04)
Treat: Late Pregnacy	0.13	0.10	0.27	0.15	0.10
	(0.05)	(0.04)	(0.07)	(0.06)	(0.04)

E. Total hours of hired labor

	Planting	Fertilizer	Weeding	Harvest	Land Clearing
Control: Non-Pregnant	4.83	3.57	34.95	18.37	6.18
	(1.29)	(1.24)	(11.60)	(4.97)	(3.18)
Control: Early Pregnacy	7.23	3.72	17.69	8.68	5.13
	(3.17)	(2.25)	(4.88)	(5.04)	(2.29)
Control: Late Pregnancy	4.55	5.27	27.62	13.02	8.53
	(2.33)	(2.90)	(11.58)	(4.47)	(4.99)
Treat: Non-Pregnant	8.74	5.96	31.84	21.68	8.52
	(2.07)	(1.18)	(5.39)	(4.69)	(2.60)
Treat: Early Pregnacy	13.29	7.19	30.90	22.55	12.15
	(5.58)	(3.13)	(7.40)	(6.90)	(4.08)
Treat: Late Pregnacy	3.30	3.18	17.67	9.35	3.78
	(1.43)	(1.81)	(7.18)	(3.78)	(1.80)

Note: Standard errors calculated via wild cluster bootstrapping.

Table A.6: Balance of characteristics between treated and controlled group among non-pregnant group

Characteristic	Control Farmers (N = 190)	Treated Farmers (N = 256)	p-value	N	Clusters
Proportion reported pregnant at time trial start in OAF data	0.25 (0.03)	0.16 (0.02)	0.313	446	64
Reported household size at trial start	4.84 (0.12)	5.30 (0.10)	0.022	446	64
Reported lines worked per day	22.98 (1.11)	27.15 (0.92)	0.408	446	64
Reported hours worked per day	4.34 (0.13)	4.16 (0.08)	0.489	446	64
Married and living together	0.84 (0.03)	0.78 (0.03)	0.060	446	64
Number of prior births	3.84 (0.12)	3.77 (0.12)	0.489	446	64
Age	32.85 (0.47)	33.60 (0.50)	0.136	446	64
Proportion reported pregnant during trial in research data	0.00 (0.00)	0.00 (0.00)	.	446	64
Total acres owned	1.75 (0.09)	1.83 (0.08)	0.869	445	64
F-test of joint significance (F-stat)		2.60	0.016	445	64

Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

Table A.7: Balance of characteristics between treated and controlled group among early pregnancy group

Characteristic	Control Farmers (N = 75)	Treated Farmers (N = 78)	p-value	N	Clusters
Proportion reported pregnant at time trial start in OAF data	0.87 (0.04)	0.99 (0.01)	0.005	153	57
Reported household size at trial start	4.20 (0.17)	4.36 (0.18)	0.504	153	57
Reported lines worked per day	22.79 (1.64)	24.15 (1.52)	0.830	153	57
Reported hours worked per day	3.85 (0.21)	3.90 (0.13)	0.970	153	57
Married and living together	0.84 (0.04)	0.95 (0.03)	0.019	153	57
Number of prior births	2.79 (0.21)	2.24 (0.18)	0.177	153	57
Age	32.17 (0.74)	30.04 (0.65)	0.029	153	57
Proportion reported pregnant during trial in research data	1.00 (0.00)	1.00 (0.00)	.	153	57
Total acres owned	1.59 (0.13)	1.80 (0.14)	0.180	153	57
F-test of joint significance (F-stat)		3.95	0.001	153	57

Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

Table A.8: Balance of characteristics between treated and controlled group among late pregnancy group

Characteristic	Control Farmers (N = 60)	Treated Farmers (N = 60)	p-value	N	Clusters
Proportion reported pregnant at time trial start in OAF data	0.92 (0.04)	0.93 (0.03)	0.729	120	56
Reported household size at trial start	4.28 (0.21)	4.52 (0.21)	0.301	120	56
Reported lines worked per day	21.15 (2.01)	25.68 (1.92)	0.348	120	56
Reported hours worked per day	4.25 (0.25)	4.06 (0.19)	0.600	120	56
Married and living together	0.78 (0.05)	0.93 (0.03)	0.019	120	56
Number of prior births	2.73 (0.23)	2.60 (0.22)	0.681	120	56
Age	32.25 (0.83)	32.80 (0.79)	0.597	120	56
Proportion reported pregnant during trial in research data	1.00 (0.00)	1.00 (0.00)	.	120	56
Total acres owned	1.52 (0.09)	1.84 (0.16)	0.037	120	56
F-test of joint significance (F-stat)		5.10	¡ 0.001	120	56

Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

Table A.9: Balance of characteristics between treated and controlled group among participants dropped from trial

Characteristic	Control Farmers (N = 433)	Treated Farmers (N = 626)	p-value	N	Clusters
Proportion reported pregnant at time trial start in OAF data	0.56 (0.02)	0.53 (0.02)	0.850	1059	70
Reported household size at trial start	4.50 (0.07)	5.06 (0.06)	0.068	1059	70
Reported lines worked per day	27.30 (0.76)	21.47 (0.54)	0.502	1059	70
Reported hours worked per day	4.70 (0.09)	4.21 (0.05)	0.440	1059	70
F-test of joint significance (F-stat)		2.33	0.065	1059	70

Note: Results presented as mean (standard error). p-values calculated via simple linear regression models with district-level fixed effects and standard errors clustered at the One Acre Fund site-level. Lines worked, hours worked, and household size winsorized at the 1st and 99th percentiles.

Table A.10: Planting Partners implementation

Metric	Eligible farmers in treated sites (N = 433)
Utilized Planting Partner program	354 (81.76%)
Total hours worked for farmer	13.62 (0.46)
Perceived PP as better farmer than self	88 (24.86%)
Perceived skill of casual	
Worst	16 (4.51%)
Poor	16 (4.51%)
Average	41 (11.55%)
Good	58 (16.34%)
Best	224 (63.10%)

Note: Results presented as N (%), save for 'Total hours worked for farmer', which is presented as mean (SE).

Table A.11: Planting Partners implementation among non-pregnant group

Metric	Eligible farmers in treated sites (N = 256)
Utilized Planting Partner program	212 (82.81%)
Total hours worked for farmer	13.27 (0.61)
Perceived PP as better farmer than self	55 (25.94%)
Perceived skill of casual	
Worst	12 (5.66%)
Poor	6 (2.83%)
Average	23 (10.85%)
Good	38 (17.92%)
Best	133 (62.74%)

Note: Results presented as N (%), save for 'Total hours worked for farmer', which is presented as mean (SE).

Table A.12: Planting Partners implementation among early pregnancy group

Metric	Eligible farmers in treated sites (N = 78)
Utilized Planting Partner program	61 (78.21%)
Total hours worked for farmer	14.26 (1.12)
Perceived PP as better farmer than self	14 (22.95%)
Perceived skill of casual	
Worst	0 (0.00%)
Poor	5 (8.20%)
Average	7 (11.48%)
Good	9 (14.75%)
Best	40 (65.57%)

Note: Results presented as N (%), save for 'Total hours worked for farmer', which is presented as mean (SE).

Table A.13: Planting Partners implementation among late pregnancy group

Metric	Eligible farmers in treated sites (N = 60)
Utilized Planting Partner program	49 (81.67%)
Total hours worked for farmer	14.53 (1.33)
Perceived PP as better farmer than self	11 (22.45%)
Perceived skill of casual	
Worst	3 (6.12%)
Poor	2 (4.08%)
Average	9 (18.37%)
Good	4 (8.16%)
Best	31 (63.27%)

Note: Results presented as N (%), save for 'Total hours worked for farmer', which is presented as mean (SE).

Table A.14: Intention to treat effects on main outcomes with no covariates

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Total agricultural yield (USD)	102.95 (64.14)	750.67 (38.72)	792	71
Maternal food security	0.09 (0.09)	0.00 (0.05)	792	71
Child solid food consumption	0.07 (0.06)	0.00 (0.05)	792	71
Average daily frequency of breastfeeding	-0.73* (0.39)	7.67 (0.29)	786	70

Note: OLS models with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Maternal food security a standardized composite measure of reports of hunger, meal frequency, and protein consumption. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.15: Intention to treat effects on main outcomes with covariates selected via double-selection-Lasso

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Total agricultural yield (USD)	63.57 (61.09)	750.67 (38.72)	791	71
Maternal food security	0.07 (0.09)	0.00 (0.05)	791	71
Child solid food consumption	0.00 (0.05)	0.00 (0.05)	791	71
Average daily frequency of breastfeeding	-0.47 (0.37)	7.67 (0.29)	785	70

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age selected via double-selection-Lasso, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.16: Intention to treat effects on individual crop yields in USD

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Maize	55.75 (62.87)	648.17 (34.32)	791	71
Soybeans	15.48* (8.90)	20.94 (3.63)	791	71
Millet	-0.03 (0.32)	0.60 (0.20)	791	71
Pigeon peas	-2.86 (2.47)	10.47 (1.39)	791	71
Groundnuts	2.95 (15.58)	48.05 (7.63)	791	71
Beans	-2.58 (3.10)	7.25 (1.87)	791	71
Tobacco	2.19 (11.67)	15.18 (5.60)	791	71

Note: OLS models adjusted for whether the farmer gave birth during the trial, total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Crop production winsorized at the 1st and 99th percentile ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.17: Intention to treat effects on food consumption

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Whether the farmer went to bed hungry a few or more times in the past month	-0.03 (0.04)	0.33 (0.02)	791	71
Whether farmer had three meals yesterday	0.06 (0.04)	0.35 (0.03)	791	71
Whether farmer had three meals the day before	0.05 (0.05)	0.34 (0.02)	791	71
Days of maternal meat consumption in the past month	-0.20 (0.24)	2.09 (0.16)	791	71
Days of maternal egg consumption in the past month	-0.29 (0.30)	2.24 (0.18)	791	71
Days of maternal usipa consumption in the past month	0.02 (0.53)	3.94 (0.29)	790	71
Days of maternal fish consumption in the past month	0.73* (0.37)	3.86 (0.24)	791	71
Days of child meat consumption in the past month	0.23 (0.49)	5.42 (0.34)	787	71
Days of child fruit & vegetable consumption in the past month	-0.08 (0.66)	12.08 (0.58)	787	71
Days of child carb consumption in the past month	-0.56 (0.57)	22.52 (0.59)	787	71

Note: OLS models adjusted for whether the farmer gave birth during the trial, total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.18: Pooled treatment on the treated effects on main outcomes

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Total agricultural yield (USD)	86.77 (74.29)	750.67 (38.72)	791	71
Maternal food security	0.09 (0.11)	0.00 (0.05)	791	71
Child solid food consumption	-0.02 (0.06)	0.00 (0.05)	791	71
Average daily frequency of breastfeeding	-0.65 (0.45)	7.67 (0.29)	785	70

Note: Models estimated via 2SLS and adjusted for whether the farmer gave birth during the trial, total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Maternal food security a standardized composite measure of reports of hunger, meal frequency, and protein consumption. Child food security is a standardized composite measure of frequency of meat; fruits and vegetables; and carbohydrate consumption.***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.19: Treatment on the treated effects on main outcomes using an Abadie's kappa-based approach

Outcome	Estimate (SE)	Control Mean (SE)	N	Clusters
Total agricultural yield (USD)	93.56 (72.76)	750.67 (38.72)	791	71
Maternal food security	0.09 (0.11)	0.00 (0.05)	791	71
Child solid food consumption	-0.02 (0.06)	0.00 (0.05)	791	71
Average daily frequency of breastfeeding	-0.57 (0.45)	7.67 (0.29)	785	70

Note: Models estimated with a covariate balancing estimator based on Abadie's kappa and adjusted for whether the farmer gave birth during the trial, total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.20: ITT effects on main outcomes by pregnancy group

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Total agricultural yield	0.02 (0.12)	0.26* (0.15)	0.06 (0.14)	718	71
Maternal food security	-0.10 (0.10)	0.39** (0.19)	0.09 (0.19)	718	71
Child solid food consumption	0.02 (0.08)	-0.24** (0.11)	0.05 (0.15)	718	71
Average daily frequency of breastfeeding	-0.17* (0.09)	-0.04 (0.12)	0.06 (0.14)	716	70

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.21: Benjamini-Hochberg q-values adjusted for multiple hypothesis testing

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy
Total agricultural yield	0.870	0.256	0.870
Maternal food security	0.793	0.249	0.870
Child solid food consumption	0.870	0.249	0.870
Average daily frequency of breastfeeding	0.256	0.870	0.870

Note: Original p-values from OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso.

Table A.22: Romano -Wolf p-values adjusted for multiple hypothesis testing

Outcome	Treat	Treatment × Early Pregnancy	Treatment × Late Pregnancy
Total agricultural yield	1.000	0.802	0.990
Maternal food security	0.802	0.109	0.762
Child solid food consumption	1.000	0.663	1.000
Average daily frequency of breastfeeding	0.465	0.990	0.465

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Adjustments calculated for main treatment exposure and indicators for the interaction between treatment and pregnancy group.

Table A.23: ITT effects on main outcomes by pregnancy group, outcomes non-standardized

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Total agricultural yield (USD)	13.95 (85.25)	189.73* (110.28)	44.28 (101.74)	718	71
Maternal food security	-0.10 (0.10)	0.39** (0.19)	0.09 (0.19)	718	71
Child solid food consumption	0.02 (0.08)	-0.24** (0.11)	0.05 (0.15)	718	71
Average daily frequency of breastfeeding	-0.91* (0.52)	-0.22 (0.66)	0.34 (0.80)	716	70

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.24: TOT effects on main outcomes among non-pregnant group

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.04 (0.13)	445	64
Maternal food security	-0.09 (0.12)	445	64
Child solid food consumption	0.02 (0.09)	445	64
Average daily frequency of breastfeeding	-0.22* (0.12)	445	64

Note: 2SLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.25: TOT effects on main outcomes among early pregnancy group

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.38* (0.19)	153	57
Maternal food security	0.52** (0.25)	153	57
Child solid food consumption	-0.29* (0.15)	153	57
Average daily frequency of breastfeeding	-0.03 (0.15)	152	56

Note: 2SLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.26: TOT effects on main outcomes among late pregnancy group

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.09 (0.17)	120	56
Maternal food security	0.08 (0.22)	120	56
Child solid food consumption	0.09 (0.19)	120	56
Average daily frequency of breastfeeding	0.12 (0.17)	119	55

Note: 2SLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.27: TOT effects on main outcomes among non-pregnant group using an Abadie's kappa-based approach

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.05 (0.13)	445	64
Maternal food security	-0.07 (0.13)	445	64
Child solid food consumption	0.04 (0.09)	445	64
Average daily frequency of breastfeeding	-0.18 (0.11)	445	64

Note: Models estimated with a covariate balancing estimator based on Abadie's kappa and adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.28: TOT effects on main outcomes among early pregnancy group using an Abadie's kappa-based approach

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.37** (0.18)	153	57
Maternal food security	0.45* (0.24)	153	57
Child solid food consumption	-0.29* (0.15)	153	57
Average daily frequency of breastfeeding	-0.01 (0.14)	152	56

Note: Models estimated with a covariate balancing estimator based on Abadie's kappa and adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.29: TOT effects on main outcomes among late pregnancy group using an Abadie's kappa-based approach

Outcome	Estimate (SE)	N	Clusters
Total agricultural yield	0.07 (0.16)	120	56
Maternal food security	0.08 (0.20)	120	56
Child solid food consumption	0.07 (0.20)	120	56
Average daily frequency of breastfeeding	0.13 (0.17)	119	55

Note: Models estimated with a covariate balancing estimator based on Abadie's kappa and adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. All outcomes standardized. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.30: ITT effects on labor allocations by pregnancy group

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Total hours of hired labor	-7.86 (15.12)	18.96 (18.08)	-34.63* (19.37)	718	71
Total hours of unpaid labor	-2.30 (1.58)	1.39 (2.90)	0.83 (3.80)	718	71
Hours of household labor	-1.08 (3.03)	-1.94 (6.13)	0.87 (9.77)	718	71
Hours of self-sourced labor	-0.97 (1.03)	0.12 (1.88)	0.55 (2.79)	718	71
Engages in non-agricultural IGAs	-0.07 (0.05)	0.06 (0.09)	0.19** (0.08)	718	71

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. Abbreviations: IGA = income-generating activity. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.31: ITT effects on likelihood of hiring labor for each activity by pregnancy group

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Planting	0.00 (0.04)	0.03 (0.07)	0.00 (0.06)	718	71
Fertilizer	-0.04 (0.04)	0.03 (0.05)	-0.04 (0.06)	718	71
Weeding	-0.08 (0.05)	0.07 (0.07)	-0.10 (0.09)	718	71
Harvest	-0.04 (0.05)	0.14* (0.08)	-0.10 (0.08)	718	71
Land Clearing	-0.01 (0.04)	0.06 (0.06)	0.01 (0.05)	718	71

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.32: ITT effects on total hours of hired labor for each activity by pregnancy group

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Planting	2.78 (2.58)	2.33 (4.91)	-1.15 (2.09)	718	71
Fertilizer	1.04 (1.81)	1.48 (2.93)	-3.35 (3.39)	718	71
Weeding	-9.35 (10.64)	4.86 (6.98)	-16.08 (11.20)	718	71
Harvest	-0.86 (5.90)	7.55 (6.52)	-10.47 (6.48)	718	71
Land Clearing	-0.24 (3.79)	4.67 (4.60)	-5.33 (4.45)	718	71

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.33: ITT effects on land allocations by pregnancy group

Outcome	Non-Pregnant	Early Pregnancy	Late Pregnancy	N	Clusters
Household rents out land	0.00 (0.03)	0.03 (0.04)	0.02 (0.02)	717	71
Household rents land for use	-0.04 (0.06)	0.13 (0.10)	0.05 (0.07)	718	71
Amount of land rented for use	0.00 (0.07)	0.21* (0.12)	0.06 (0.09)	718	71
Proportion of household land cultivated	-0.03* (0.02)	-0.02 (0.02)	-0.05** (0.03)	691	70

Note: OLS models adjusted for total acres owned by the household, household size, marital status, and age, with district-level fixed effects and standard error clustered at the One Acre Fund site-level. Interactions between these covariates and pregnancy status were additionally selected via double-selection-Lasso. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A.34: History of pregnancy during the agricultural season among the non-pregnant group

Group	Ever Early Pregnacy	Previous Early Pregnacy	Ever Late Pregnacy	Previous Late Pregnacy	Ever Either	Previous Either
Treat	84.77% (2.25)	37.50% (3.03)	55.47% (3.11)	16.41% (2.32)	96.09% (1.21)	53.91% (3.12)
Control	86.84% (2.46)	47.89% (3.63)	55.26% (3.62)	13.68% (2.50)	95.26% (1.55)	61.58% (3.54)
Both	85.65% (1.66)	41.93% (2.34)	55.38% (2.36)	15.25% (1.70)	95.74% (0.96)	57.17% (2.35)

Note: Prevalence % (standard error) of non-pregnant group with a prior birth that would have categorized them in the early or late pregnancy groups, ever or in the previous agricultural season.

Appendix B Supplementary Model Exhibits

Figure B.1: Pregnancy exposure and total effective on-farm labor supply

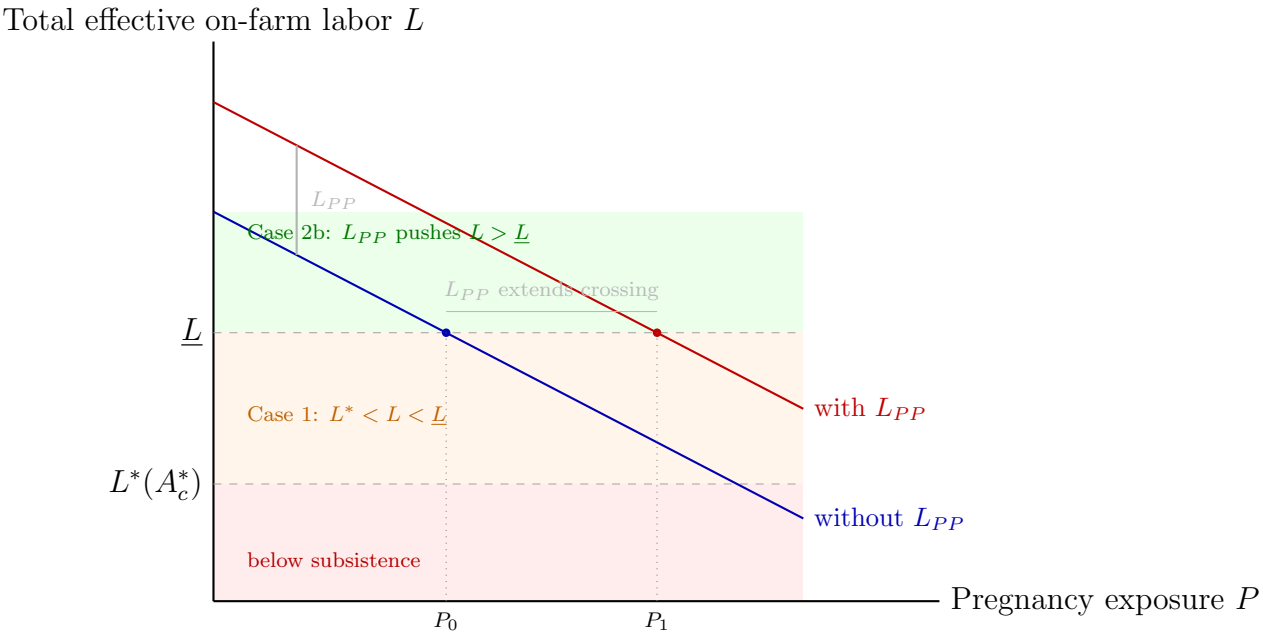
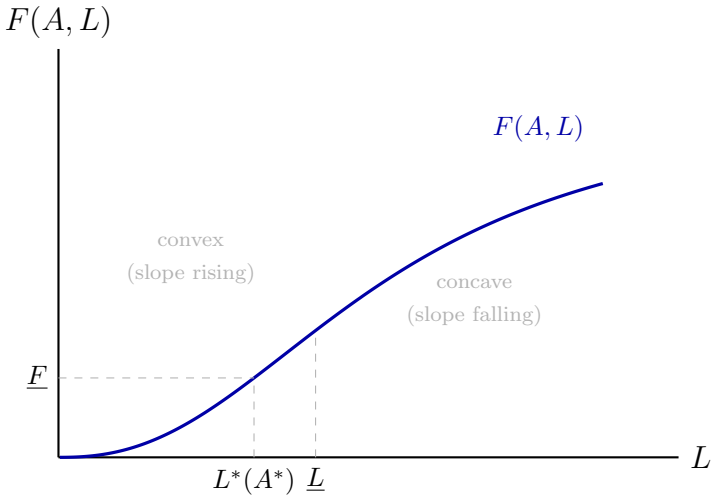
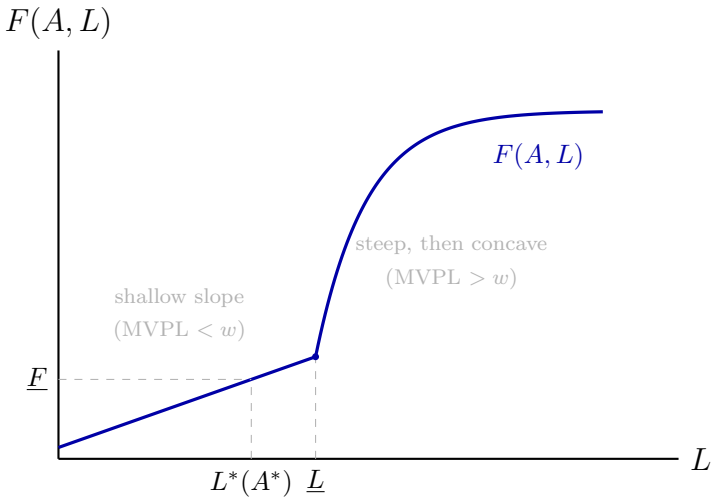


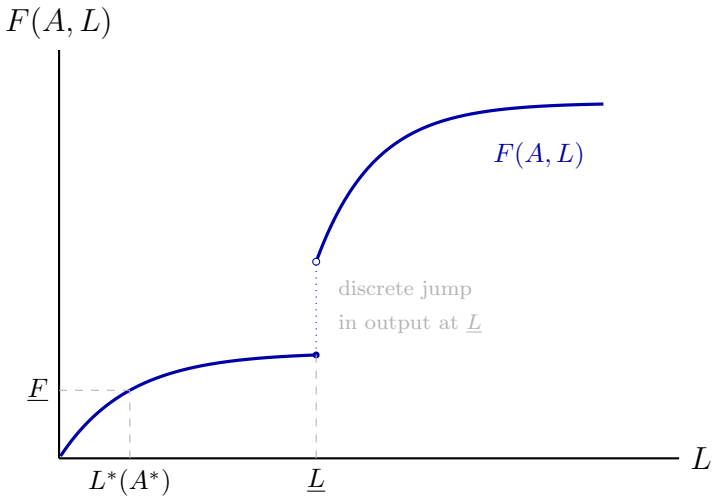
Figure B.2: Total effective on-farm labor and farm production



(1) Smooth S-shape



(2) Kinked at $\underline{L}$



(3) Discontinuous jump

B.1 Land and labor responses to Planting Partners: with or without complete markets

Table B.1: Testable implications of incomplete markets, by labor regime

Regime	**Interpretation**	**Testable condition**	**Graph**
position of $L = \phi_f(P)L_w + \bar{L}_o^N + \bar{L}_h + L_{PP}$ relative to $L^*(A_c^*)$ and $\underline{L}$			
Case 1 $L^*(A_c^*) < L < \underline{L}$	Households still choose $L = L^*(A_c^*)$, but now L_{PP} frees labor (household or hired) from subsistence farming, allowing off-farm labor or leisure rise, especially at higher P. *Necessary, not sufficient* conditions for incomplete markets.	$\frac{\partial^2(\ell_w^* + z_w^* + \ell_o^* + z_o^*)}{\partial L_{PP}\partial P} > 0$ and $\frac{\partial^2(L_w^* + L_o^* + L_h^*)}{\partial L_{PP}\partial P} < 0$	Figure 5 Panel (b)
Case 2(a) $\underline{L} < L$	Land or output responds differentially to L_{PP} by pregnancy; separation fails. *Necessary and sufficient* conditions for incomplete markets.	$\frac{\partial^2(A_r^* + A_c^*)}{\partial L_{PP}\partial P} > 0$ or $\frac{\partial^2 Y}{\partial L_{PP}\partial P} > 0$	
	If no market substitutes for women's labor, none is crowded out. *Sufficient (not necessary)* for zero substitution.	$\frac{\partial^2 A_r^*}{\partial L_{PP}\partial P} \geq 0$, $\frac{\partial^2 L_h^*}{\partial L_{PP}\partial P} \geq 0$, $\frac{\partial^2 L_o^*}{\partial L_{PP}\partial P} \geq 0$	
Case 2(b) $L - L_{PP} < \underline{L} < L$	All of 2(a); *plus* crossing $\underline{L}$ raises the marginal product of on-farm labor above w. If there is slack in $\bar{L}_h$ or $\bar{L}_o^N$, on-farm labor rises.	$\frac{\partial(L_h^* + L_o^*)}{\partial L_{PP}} \geq 0$	Figure 5 Panel (c)

B.1.1 Case 1: $L^*(A_c^*) < \phi_f(P)L_w + \bar{L}_o^N + \bar{L}_h + L_{PP} < \underline{L}$

This case is represented stylistically by Figure 5 Panel (b). Even with planting partners labor hours, the household's total effective on-farm labor is still below the threshold $\underline{L}$. However, with the planting partners labor hours, the household is above the minimal labor for subsistence consumption $L^*(A_c^*)$.

Necessary Conditions for Incomplete Markets:

$$\frac{\partial^2(\ell_w^* + z_w^* + \ell_o^* + z_o^*)}{\partial L_{PP}\partial P} > 0 \quad \text{and} \quad \frac{\partial^2(L_w^* + L_o^* + L_h^*)}{\partial L_{PP}\partial P} < 0 \tag{7}$$

When total effective farm labor supply is below the threshold $\underline{L}$, then households should only farm enough to produce subsistence production $\underline{F}$, and should dedicate their remaining labor hours off-farm. When markets are incomplete, pregnant women must still

provide labor on-farm in order to ensure that the household meets subsistence. If pregnant women are differentially more likely to move labor hours off-farm when provided with L_{PP}, it implies that they are not simply substituting away from the labor that they optimally supplied to the farm (a marker of complete markets, as depicted in the lower panel of Figure 5 Panel (a)).

B.1.2 Case 2(a): $\underline{L} < \phi_f(P)L_w + \bar{L}_o^N + \bar{L}_h + L_{PP}$

With planting partners labor hours L_{PP}, the household's total effective on-farm labor is above the threshold $\underline{L}$.

Necessary and Sufficient Conditions for Incomplete Markets:

$$\frac{\partial^2(A_r^* + A_c^*)}{\partial L_{PP}\partial P} > 0 \quad \text{or} \quad \frac{\partial^2 Y}{\partial L_{PP}\partial P} > 0 \tag{8}$$

When markets are incomplete, pregnant women either must let their land fallow or attempt to supply their own labor to the farm unproductively. This slack in the amount of land that is used productively implies that a new source of labor must lead to a total increase in the amount of land that is cultivated, or the productivity of the cultivated land, more so for pregnant women than non-pregnant women (Equation 8).

Land and Labor Allocation Conditions (Sufficient but not Necessary):

$$\frac{\partial^2 A_r^*}{\partial L_{PP}\partial P} \geq 0, \quad \frac{\partial^2 L_h^*}{\partial L_{PP}\partial P} \geq 0, \quad \text{and} \quad \frac{\partial^2 L_o^*}{\partial L_{PP}\partial P} \geq 0 \tag{9}$$

If there is no scope for any amount of substitution for pregnant women's labor supply, then Planting Partners will not substitute from any existing market or intra-household solution, giving rise to all three conditions in Equation 9.[35] While this is a sufficient condition to demonstrate that none of these markets substitute for women's labor supply while they are pregnant, it is possible that women are able to use these markets for some degree of substitution (in which case not all three conditions will be met), but not enough to offset their productivity losses (implying that markets are still incomplete). Thus, the condition in Equation 9 is sufficient but not necessary.

B.1.3 Case 2(b): $\phi_f(P)L_w + \bar{L}_o^N + \bar{L}_h < \underline{L} < \phi_f(P)L_w + \bar{L}_o^N + \bar{L}_h + L_{PP}$

This case is represented stylistically by Figure 5 Panel (c). In this case, households are below the threshold $\underline{L}$ without planting partners labor, and are pushed above it through

[35] Each second derivative condition will be greater than zero if any input is complementary to Planting Partners labor inputs, and equal to zero otherwise.

the intervention. In this case, all of the conditions in Case 2(a) still apply. However, households also experience an increase in the marginal product of labor such that it exceeds the prevailing wage w. In this regime, frictions in land markets, labor markets, and intra-household substitution lead both land *and* labor to be less productive inputs. Then, if either $L_h < \bar{L}_h$ or $L_o < \bar{L}_o^N$, we will also see an increase in on-farm labor supply:[36]

$$\frac{\partial(L_h^* + L_o^*)}{\partial L_{PP}} \geq 0$$

[36]Recall that it is possible that both constraints bind, but that, when total effective labor supply available is below $\underline{L}$, households choose $L_h < \bar{L}_h$ and $L_o < \bar{L}_o^N$. If a household is $\bar{L}_h + \epsilon$ below $\bar{L}$, she will choose to purchase no hired labor, even if she is able to purchase up to $\bar{L}_h$.